MILITARY SPECIFICATION

COMMUNICATIONS SYSTEMS INTEGRATED VOICE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, and test requirements for a computer-controlled, automatic switching service for shipboard interior voice communications and distribution, henceforth referred to as the Integrated Voice Communications System (IVCS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-5	- Capacitor, Fixed, Mica-dielectric, Style Cm 15.
MIL-R-19	- Resistor, Variable, Wirewound (Low Operating Temperature) General Specification For.
MIL-C-20	- Capacitor, Fixed, Ceramic Dielectric (Temperature Compensating), Established and Non-established Reliability, General Specification For.
MIL-R-22	- Resistor, Variable (Wirewound, Power Type), General Specification For.
MIL-R-26	- Resistor, Fixed, Wire-wound (Power Type), General Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3 Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-81 - Capacitor, Variable, Ceramic Dielectric General Specification For.
MIL-R-94 - Resistor, Variable, Composition, General Specification For.
MIL-S-901 - Shock Test, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For (Navy).
MIL-C-915 - Cable and Cord Electrical, For Shipboard Use General Specifications For.
MIL-C-915/29 - Cable, Electrical, 1000 Volts, Type DSGU.
MIL-C-915/30 - Cable, Electrical, 1000 Volts, Type TSGU.
MIL-C-915/49 - Cable, Electrical, Type 2SWU.
MIL-C-5015 - Connector, Electrical, Circular Threaded, AN Type, General Specification For.
MIL-E-6051 - Electromagnetic Compatibility Requirements, Systems.
MIL-P-7788 - Panel, Information, Integrally Illuminated.
MIL-C-10544 - Connectors, Plug and Receptacle (Electrical, Audio, Waterproof, Ten Contact, Polarized).
MIL-C-10950 - Capacitor, Fixed, Mica Dielectric, Button Style, General Specification For.
MIL-C-11015 - Capacitor, Fixed, Ceramic Dielectric (General Purpose) General Specification For.
MIL-R-11804 - Resistor, Fixed, Film (Power Type), General Specification For.
MIL-C-12889 - Capacitor, By-pass, Radio-interference Reduction, Paper Dielectric, Ac and Dc, (Hermetically Sealed In Metallic Cases) General Specification For.
MIL-R-12934 - Resistor, Variable, Wirewound, Precision General Specification For.
MIL-C-14409 - Capacitor, Variable (Piston Type, Tubular Trimmer), General Specification For.
MIL-E-15090 - Enamel, Equipment, Light-gray (Formula No. 111).
MIL-T-15514 - Telephone Equipment, Sound Powered Handsets, Headset-Chest Sets, and Headset-Microphones, Types H-200/U, H-201/U, H-202/U, H-203/U, and H-204/U.
MIL-F-15733 - Filter, Radio Interference, General Specification For.
MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification For.
MIL-W-16878 - Wire, Electrical, Insulated, General Specification For.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of.
MIL-F-18327 - Filter, High Pass, Low Pass, Band Pass, Band Suppression and Dual Functioning, General Specification For.
MIL-R-18546 - Resistor, Fixed, Wirewound (Power Type, Chassis Mounted) General Specification For.

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MIL-S-19500 - Semiconductor Device, General Specification For.
MIL-C-19978 - Capacitor, Fixed, Plastic (or Paper-plastic) Dielectric, (Hermetically Sealed In Metal, Ceramic, or Glass Cases) Established and Nonestablished Reliability, General Specification For.
MIL-R-22097 - Resistor, Variable, Nonwire Wound (Adjustment Types), General Specification For.
MIL-C-23183 - Capacitor, Fixed or Variable, Vacuum Dielectric, General Specification For.
MIL-C-23269 - Capacitor, Fixed, Glass Dielectric, Established Reliability, General Specification For.
MIL-R-23285 - Resistor, Variable, Nonwire-wound General Specification For.
MIL-R-27208 - Resistor, Variable, Wire-wound (Adjustment Type), General Specification For.
MIL-C-28840 - Connector, Electrical, Circular Threaded, High Density, High Shock Shipboard, Class D.
MIL-M-38510 - Microcircuit, General Specification For.
MIL-C-39001 - Capacitor, Fixed, Mica Dielectric, Established Reliability, General Specification For.
MIL-R-39002 - Resistor, Variable, Wirewound, Semi-precision, General Specification For.
MIL-C-39003 - Capacitor, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Established Reliability, General Specification For.
MIL-R-39005 - Resistor, Fixed, Wirewound (Accurate), Established Reliability, General Specification For.
MIL-C-39006 - Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte) Tantalum, Established Reliability, General Specification For.
MIL-R-39007 - Resistor, Fixed, Wirewound (Power Type), Established Reliability, General Specification For.
MIL-R-39008 - Resistor, Fixed, Composition (Insulated) Established Reliability, General Specification For.
MIL-R-39009 - Resistor, Fixed, Wirewound (Power Type, Chassis Mounted), Established Reliability, General Specification For.
MIL-C-39014 - Capacitor, Fixed, Ceramic Dielectric (General Purpose) Established Reliability, General Specification For.
MIL-R-39015 - Resistor, Variable, Wire-wound (Lead Screw Actuated), Established Reliability, General Specification For.
MIL-R-39017 - Resistor, Fixed, Film, (Insulated) Established Reliability, General Specification For.
MIL-C-39018 - Capacitor, Fixed, Electrolytic (Aluminum Oxide), Established Reliability and Non-established Reliability, General Specification For.
MIL-C-39022 - Capacitor, Fixed, Metallized Paper, Paper-plastic Firm, Or-plastic Film Dielectric, Direct and Alternating Current, (Hermetically Sealed In Metal Cases), Established Reliability, General Specification For.

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MIL-R-39023 - Resistor, Variable, Non Wire Wound, Precision, General Specification For.
MIL-R-39035 - Resistor, Variable, Nonwire Wound (Adjustment Type), Established Reliability, General Specification For.
MIL-I-46058 - Insulating Compound, Electrical (For Coating Printed Circuit Assemblies).
MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw and Stud Types, and Associated Accessories, General Specification For.
MIL-R-55182 - Resistor, Fixed, Film, Established Reliability, General Specification For.
MIL-C-55365 - Capacitor, Chip, Fixed, Tantalum, Established Reliability, General Specification For.
MIL-C-55681 - Capacitor, Chip, Multiple Layer, Fixed Unencapsulated, Ceramic Dielectric, Established Reliability, General Specification For.
MIL-W-81044 - Wire, Electric, Crosslinked Polyalkene, Crosslinked Alkane-imide Polymer, or Polyarylene Insulated, Copper or Copper Alloy.

STANDARDS

FEDERAL

FED-STD-595 - Color.

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements For Enclosure For Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-242 - Electronic Equipment Parts (Selected Standards).
MIL-STD-415 - Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-471 - Maintainability/Verification/Demonstration/Evaluation.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests Exponential Distribution.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

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MIL-STD-965 - Parts Control Program.
MIL-STD-1132 - Switch and Associated Hardware, Selection and Use of.
MIL-STD-1277 - Splices, Chips, Terminals, Terminal Boards, Binding Posts, Electrical.
MIL-STD-1286 - Transformer, Inductors, and Coils Selection and Use of.
MIL-STD-1310 - Shipboard Bonding, Grounding and Other Techniques For Electromagnetic Compatibility and Safety.
MIL-STD-1346 - Relay, Selection and Application.
MIL-STD-1353 - Electrical Connectors, Plug In Sockets and Associated Hardware, Selection and Use of.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems Electric Power, Alternating Current (Metric).
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1680 - Installation Criteria for Shipboard Secure Electrical Information Processing Systems (U).
DOD-STD-1686 - Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric).

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.
DOD-HDBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric).
MIL-HDBK-472 - Maintainability Prediction.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein.

FEDERAL COMMUNICATIONS COMMISSION (FCC)

Part 68 FCC Rules and Regulations (Telephone Equipment Requisition Program)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL SECURITY AGENCY (NSA)

NACSEM 5100 Series - National Security Agency, Compromising Emanations Laboratory Test Standard Electromagnetics. (U)

(Application for copies should be addressed to Headquarters Naval Security Group Command, Attention: Code G65, 3801 Nebraska Avenue, NW, Washington, DC 20390).

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NAVAL MATERIAL COMMAND (NAVMAT)
NAVMAT P-9492 - Navy Manufacturing Screening Program.

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS 0967-LP-312-8010 - Maintainability Design Criteria Handbook for Designers Electronic Equipment, Shipboard.
NAVSHIPS 0967-LP-316-8010 - Reliability Design Handbook of Bureau of Ships.
NAVSEA 0967-LP-597-1011 - Parts Application and Reliability Information Manual for Navy Electronic Equipment.

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ELECTRONIC INDUSTRIES ASSOCIATION
EIA-RS232C - Interface Between Data Terminal Equipment and Data Communication Equipment Employing Binary Data Interchange. (DoD adopted)

(Application for copies should be addressed to the Electronic Industries Association, Engineering Department, 2001 Eye Street, NW, Washington, DC 20006.)

RADIO CORPORATION OF AMERICA
RCA Drawing 856-0003 - Specification Navy Tactical Data Interphone System (IADS) Interface.

(Copies of this drawing required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order or precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the computer controlled, automatic switching equipment shall be subjected to first article inspection (see 4.4.1 and 6.3).

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3.2 System characterizations. The IVCS shall be a shipboard voice distribution system capable of providing both administrative and tactical communications on a 24 hour basis. The IVCS shall provide computer controlled, automatic switching service (line-to-line, line-to-net, and line-to-external interface connections), netting service, and subscriber service features to include but not be limited to the following:

(a) Four-digit dialing.
(b) Abbreviated addressing.
(c) Preset conferencing.
(d) Progressive conferencing.
(e) Call hold.
(f) Call forwarding.
(g) Call transfer.

3.3 System configuration. The IVCS shall accommodate the voice switching requirements of various ship classes providing switching services for 200 to 3200 switchable terminations. The IVCS is sized in terms of "terminations" rather than "loops", "trunks", or "lines". A loop is a single analog or digital channel between two switches. A line is a term which implies either a "loop" or "trunk". A switchable termination is described as a system port capable of providing service to a line, net, or interface. System configurations shall consist of single or multiple switching centers. A net is described as an audio bus providing common talk-common listen capability. Systems requiring 200 to 400 switchable terminations shall be configured as a single switching center. Systems requiring 400 to 1600 switchable terminations shall be configurable as either a single switching center or dual switching centers. Systems requiring 1600 to 3200 switchable terminations shall be configured as dual switching centers. Each center shall be functionally identical and capable of independent operation. Sufficient redundancy shall be provided in each switching center to meet the availability criterion specified herein.

3.4 System modularity. The system shall be of modular design, both in hardware and software components. The hardware modularity shall provide for system configurations as specified in 3.3 with a minimum number of different modules or subassemblies. The design shall include modular components for processor memory, switching matrices, main distribution frame (MDF), and terminations. Modularity shall be achieved through optimum use of microelectronics and miniature design techniques to allow for minimization of equipment size, weight, and power consumption. Modular units shall be configured to allow mounting in standard 19 inch racks and cabinets not to exceed 24 inches in width.

3.5 System description. The IVCS shall generally be comprised of the subsystems listed below:

(a) Control.
(b) Switching.
(c) Power supply and power distribution (PS/PDS).
(d) Distribution.

Subsystems which provide connectivity for system subscribers shall be interconnected by a combination of trunks, net interties, and control lines.

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3.5.1 IVCS switching center. For the purposes of this specification, IVCS subsystems assembled as an operating entity will be referred to as a switching center.

3.5.2 System performance. The IVCS shall perform as specified herein while exposed to the environmental conditions (see 3.8.9).

3.5.2.1 Capacity. The IVCS is sized in terms of the switchable termination. The switchable termination may be used in conjunction with a line, net, or interface (see 3.5.2.2), depending on the associated termination circuitry. The modular design of the IVCS shall accommodate variations in system size by changes in the number of matrices, MDF components, and terminations.

3.5.2.2 Connectivity. The IVCS shall provide switched service among terminals and shall provide interconnection between switching centers when dual centers are required.

3.5.2.2.1 Interconnections. The following types and quantities of interconnections shall be provided between switching centers in a dual center installation (see 6.2.1):

3.5.2.2.1.1 Trunks. The trunks shall provide audio paths between switching centers. The number of trunks for each configuration shall be sufficient to provide the grade of service specified in 3.5.2.3. Expansion capability (spares) shall be provided in trunks and switching center capacity equal to 15 percent of the required service. The spare trunk (cable) capacity will be provided as ship-provided cable. The spare capacity in each switching center shall be in the form of prewired spaces necessary for any printed wiring assemblies (PWAs) required to support the expansion (spare) capacity. The PWAs necessary to provide the spare capacity need not be included as part of the furnished system.

3.5.2.2.1.2 Net interties. The net interties shall provide audio paths between switching centers. The number of net interties for each configuration shall be sufficient to provide the grade of service specified in 3.5.2.3. Expansion capability (spares) shall be provided in trunks and switching center capacity equal to 15 percent of the required service. The spare cable capacity will be provided as ship-provided cable. The spare capacity in each switching center shall be in the form of prewired spaces necessary for any printed wiring assemblies (PWAs) required to support the expansion (spare) capacity. The PWAs necessary to provide the spare capacity need not be included as part of the furnished system.

3.5.2.2.1.3 Control lines. The control lines shall provide the electrical connections for control signals between switching centers and for monitoring remote center status and fault alarms.

3.5.2.2.1.4 Power lines. The power lines shall provide the means to transmit power between switching centers. The current-carrying capacity of the power lines will be sufficient to provide the total power requirements of the remote switching center and its connected terminals during emergency power losses.

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3.5.2.2.2 Local connections. The following types of connections shall be provided to terminals, nets, and local interfaces from a switching center in either a single or dual center installation.

3.5.2.2.2.1 Line connection. A line connection is the electrical connection between a switching center termination and dial terminal(s).

3.5.2.2.2.2 Net connection. A net connection is an electric circuit connecting a number of call net terminals or jackbox terminals to a termination in a switching center.

3.5.2.2.2.3 Interface connection. An interface connection is the electrical connection between a termination in a switching center and a shore line or an input or output termination of one of the ship's systems specified in 3.6.10.

3.5.2.3 Traffic handling. The IVCS is intended to be a voice distribution system providing switching service, netting service, and certain subscriber service features. Accordingly, the IVCS shall accommodate the traffic loads associated with these applications. Performance requirements of 3.5.2.1 through 3.5.2.2.2 shall be satisfied for the range of traffic loads and distributions specified in 3.5.2.3.1 through 3.5.2.3.6.

3.5.2.3.1 Traffic load and distribution assumptions. During peak traffic hours, the IVCS shall handle 0.004 line-to-line calls originated by dial terminal a second with an average holding time of 30 seconds and 0.002 line-to-net calls originated by dial terminal and by call net terminal a second with an average holding time of 2 minutes. Average holding time is exclusive of call-initiation delays, dialing, and call-completion delays. In the event there is more than one switching center, it shall be assumed that 2/3 of these calls are initiated or completed within the originating center and 1/3 in the remote center, that the traffic load imbalance between switches shall not exceed 1.5 to 1, and that the average receiver/sender (R/S) holding time shall not exceed 4 seconds a call.

3.5.2.3.2 Call busy factor adjustments. A call-busy factor of 25 percent shall be assumed to reflect the number of dial terminals unable to make or receive calls because the terminal is busy with a previously established call, for example one out of four terminals will be in use at any given time.

3.5.2.3.3 Blocking. The center shall provide a traffic-handling capability of less than one call in 1000 lost, blocked, or misrouted due to:

(a) Unavailability of interswitching center trunk.
(b) Error in R/S or control subsystem number translation or recognition.
(c) False trunk, switch, or station signal.

3.5.2.3.4 Call initiation delay. During peak traffic hours call initiation delay measured from call initiation (terminal off hook) to receipt of dial tone shall be less than 3 seconds.

3.5.2.3.5 Call completion delay. Calls that are not blocked because of traffic congestion shall be processed within an average delay of less than 0.5 second during peak traffic hours. This shall be measured from recognition of

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receipt of final address digit until initiation of signaling to the next switching center or until the start of ring. In the case of an intercenter call, an additional 2-second delay will be tolerated during peak traffic hours until start of ring.

3.5.2.3.6 Misrouting. The probability of call misrouting caused by internal switch processing, errors in reception of signaling, or errors in trunk signaling shall be less than 1 in 10^6.

3.5.2.4 Transmission. The IVCS shall meet the transmission criteria specified in 3.5.2.4.1 through 3.5.2.4.3.

3.5.2.4.1 Intraswitching center. The transmission criteria listed in table I shall be met for any termination to any other termination (except terminations used for system control) in the same switching center, including the termination circuitry, as measured at the MDF.

TABLE I. Intraswitching center transmission criteria.

Nominal test tone level	0.778 Vrms at 1 kHz
Frequency range	300 Hz to 3.3 kHz
Frequency response 1/	± 1 dB
Longitudinal balance	greater than 40 dB
Delay	
Absolute	200 microseconds maximum
Differential 2/	20 microseconds maximum
Crosstalk isolation	greater than 83 dB (design goal - greater than 100 dB)
Impulse noise	-50 dBm maximum 3/
Power line interference	-55 dB maximum from test tone level
Harmonic distortion	5 percent maximum
Signal-to-noise ratio	35 dB minimum

1/ Frequency response between 300 hertz (Hz) and 3.3 kilohertz (kHz), as measured at the MDF, shall be within plus or minus 1 dB relative to a 1 kHz reference signal.
2/ Difference in delay between any two frequencies within the range specified.
3/ Decibel unit of power (referenced to 1 milliwatt).

3.5.2.4.2 Net. The typical transmission criteria between net terminals on the same net is shown in table II. The connection of a switching center to a net shall not degrade the transmissions on the net.

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TABLE II. Net transmission criteria.

Nominal test tone level	0.778 Vrms at 1 kHz
Frequency range	300 Hz to 6 kHz
Frequency response 1/	± 1 dB
Longitudinal balance	greater than 40 dB
Insertion loss 2 Terminals 20 Terminals 30 Terminals	 1 dB ± 0.5 dB 3 dB ± 1.5 dB 6 dB ± 2.0 dB
Delay Absolute Differential 2/	 200 microseconds maximum Terminal to 20 microseconds maximum terminal
Crosstalk isolation	greater than 80 dB
Impulse noise	-50 dBm maximum 3/
Power line interference	-55 dB maximum from test tone level
Harmonic distortion	5 percent maximum
Signal-to-noise ratio	35 dB minimum

1/ Frequency response between 300 Hz and 6 kHz, as measured between net terminals, shall be within plus or minus 1 dB relative to a 1 kHz reference signal.
2/ Difference in delay between any two frequencies within the range specified.
3/ Decibel unit of power (referenced to 1 milliwatt).

3.5.2.4.3 Cabling. Table III identifies the cabling required to support and interconnect the IVCS system.

TABLE III. Required cabling.

From	To	Cable type	Maximum cable length
Dial/net terminals (via a junction box)	MDF	2SWU-19 (MIL-C-915/49)	1000 feet 1/
Center (MDF)	Center (MDF) (trunks)	2SWU-19 (MIL-C-915/49)	500 feet
Center (MDF)	Center (MDF) (net interties)	2SWU-19 (MIL-C-915/49)	500 feet

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TABLE III. Required cabling. - Continued

From	To	Cable type	Maximum cable length
Ships radio	MDF	2SWU-19 (MIL-C-915/49)	1000 feet
Public address	MDF	2SWU-19 (MIL-C-915/49)	1000 feet
Sound-powered	MDF	2SWU-19 (MIL-C-915/49)	1000 feet
MOM base station	MDF	2SWU-19 (MIL-C-915/49)	1000 feet
IADS	MDF	2SWU-19 (MIL-C-915/49)	1000 feet
Shore line control terminal (SLCT)	MDF	2SWU-19 (MIL-C-915/49)	1000 feet
Shore line communication panel(s)	MDF	2SWU-19 (MIL-C-915/49)	500 feet
Center (PSC)	Center (PSC) (control and indicators)	2SWU-19 (MIL-C-915/49)	500 feet
Center (D.c. breaker panel)	Center (D.c. breaker panel) (power and return)		
	24 V	(2)DSGU-75 (MIL-C-915/29)	500 feet (each)
	-48 V	DSGU-75 (MIL-C-915/29)	500 feet
Ship power	PSC		
	440 V 3 Phase	TSGU-14 (MIL-C-915/30)	
	115 V 1 Phase	DSGU-9 (MIL-C-915/29)	
Battery	D.c. power breaker panel		
	24 V	TSGU-100 (MIL-C-915/30)	
	-48 V	DSGU-75 (MIL-C-915/29)	
D.c. power breaker panel	PSC	TSGU-100 (MIL-C-915/30) DSGU-75 (MIL-C-915/29)	

1/ 2500 feet for dial terminal to dial terminal intercenter trunk calls.

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3.6 Subsystem.

3.6.1 Power supply and power distribution subsystem.

3.6.1.1 Description. The subsystem shall provide the required alternating current (a.c.) or direct current (d.c.) power for all other subsystems of the IVCS and the distribution of that power to those subsystems. Input power for the subsystem will be supplied from ships power or batteries.

3.6.1.2 Performance. The subsystem shall operate within the tolerances specified in MIL-E-16400 and shall be grounded in accordance with the requirements of MIL-STD-1310 and MIL-STD-1680. Redundant power supplies shall be provided as required to maintain system reliability. Automatic transfer to batteries on loss of ship's power shall be provided with no interruption or degradation of calls in progress or other operating characteristics of the IVCS. The subsystem shall provide automatic fault isolation and short indications in the power supplies or batteries. Power supplies may be either modular or in discrete sizes to meet the requirement of the switching system.

3.6.1.2.1 Power inputs. The IVCS PS/PDS will be supplied with 440 volts alternating current (Va.c.), 3 phase, 60 Hz power and 115 Va.c., single phase, 60 Hz power from the ship's power. The input power will have the standard electrical a.c. shipboard characteristics as specified for type I power of DOD-STD-1399, section 300.

3.6.1.2.2 Power outputs. The subsystem shall provide all of the power required by the system. Examples include:

(a) 115 Va.c., single phase, 60 Hz to peripherals as required.
(b) +24 Vd.c. to:

(1) Terminals through the distribution subsystem.
(2) Remote switching centers.
(3) Batteries.

(c) Other d.c. voltage as required.

3.6.1.3 A.c. power. The a.c. power is provided by shipboard power sources to the interior communication (IC) switchboards. The IC switchboard provides a.c. power for the IVCS. A separate 115 Va.c. power connection will be provided for each peripheral. A visual indicator shall be provided for each separate power input.

3.6.1.4 D.c. power breaker panel. The d.c. power breaker panel shall be used to connect and disconnect the switching center batteries to and from the center +24 Vd.c. and other d.c. power supplies as required. The d.c. breakers shall conform to the requirements of MIL-STD-242.

3.6.1.5 D.c. power supplies. The d.c. power supplies shall accept ship-provided a.c. power as specified in 3.6.1.2.1 and convert it to +24 Vd.c. and other d.c. voltage as required for distribution as specified in 3.6.1.2.2. In single center configurations, there shall be two +24 Vd.c. power supplies. In dual center configurations, there shall be one +24 Vd.c. power supply in each switching center. Each d.c. power supply of each center shall be capable of

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providing sufficient power to operate two fully populated centers and associated terminals. It shall be possible to supply +24 Vd.c. and other d.c. as required from a remote center to a center which has lost power. Each d.c. power supply shall also be used for charging the associated emergency battery power supply. Other d.c. power supplies will be provided as required.

3.6.1.6 Batteries and racks. There shall be a +24 Vd.c. battery power supply in each switching center to be provided separately. Other d.c. battery power supplies will be provided as required. The batteries shall be capable of supplying required power to the center and associated terminals for 1 hour at 25 degrees Celsius (°C). At 0°C the maximum allowable reduction in output shall be 10 percent.

3.6.2 Control subsystem.

3.6.2.1 Description. The control subsystem shall provide the control and monitoring functions, including status indications, necessary to provide the automatic switching and subscriber services available within the IVCS. In addition, this subsystem shall control maintenance and diagnostic functions. The control subsystem in each switching center shall consist of two processors, each with associated memory, a switchover control unit (SCU), and control programs. Typical peripheral devices utilized with this subsystem are identified in 3.6.5.

3.6.2.2 Performance.

3.6.2.2.1 General. The control subsystem shall respond to manual controls and to the signaled requests for service by assembling the system resources in the appropriate sequence to accommodate each requirement and reallocating the system resources when each requirement is complete. The management of system resources shall include all sequential actions taken in response to: manually initiated or automatic system interrupts, terminal requests, and the interaction with peripherals. In addition, the control subsystem will perform the following functions:

(a) Timing and counting functions including length of tones, length of dialing, and number of digits dialed.
(b) Signaling tone validation including examination of the type and identity of keyed digits.
(c) Number translation including correlation of a dialed number to an equipment location including abbreviated address, nets, call net terminals, remote numbers, and dual tone multiple frequency (DTMF) to dial pulse dial pulse to DTMF conversion.
(d) On-line fault detection and status of all critical hardware or components.
(e) Validation of individual terminal permits and accesses.

3.6.2.2.2 Controls and indicators. The control subsystem shall provide the system controls and system status indicators identified in tables IV and V as a minimum. Visual indicators shall conform to the requirements of MIL-STD-1472.

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TABLE IV. System status panel controls.

Control	Function
Fault reset	Resets all fault conditions in the SCU and extinguishes all Fault mode lamps displayed on the system status panel.
Lamp test	Lights all lamp indicators and push switch indicators for testing.
On-line select	Provides manual or automatic selection of on-line Processor A or B.
Switchover	Enables the SCU to automatically monitor and control the synchronization of processors (RESYNC) or remove the SCU from control of the processors (Comp off).
Select printer	Provides manual or automatic assignment of printer to Processor A or B.
Tape reader select	Provides manual or automatic assignment of tape reader to on-line processor manual assigned to Processor A or B.
Memory write	Allows modification of software in either or both memories when in the ON position.
Load	Commands the magnetic tape drive to rewind the tape to load point.

TABLE V. System status panel indicator.

Indicator	Type indicator	Function
Matrix on-line	Visual	Indicates matrix is currently providing IVCS switching center connections.
Matrix fault	Visual	Indicates a failure in matrix.
Processor A/B on-line	Visual	Indicates which processor is controlling the IVCS switching center.
Processor A/B fault	Visual	Indicates a failure in Processor A or B.

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TABLE V. System status panel indicator. - Continued

Indicator	Type indicator	Function
Processor A/B printer control	Visual	Indicates the processor which has control of the printer.
Switchover control Unit On mode	Visual	Indicates the SCU is in the automatic position.
Switchover control Unit Fault	Visual/ audible alarm	Indicates a failure in the SCU.
Switchover control Unit Comp off	Visual	Indicates the SCU is not controlling the synchronization of the processors.
Temperature	Visual/ audible alarm	Indicates an overtemperature condition within the system.
Fuse	Visual/ audible alarm	Indicates a fuse failure in a terminal line circuit.
Power	Visual/ audible alarm	Indicates a power failure within the system.
Power control	Visual/ audible alarm	Indicates a failure within the power distribution system.
On-line	Visual	Tape unit has power on, cartridge in place, and is not in manual mode.
Busy	Visual	Tape drive is running.
Load point	Visual	Tape is positioned to read first block of data on tape.

3.6.2.3 Processor.

3.6.2.3.1 Description. The processors of the control subsystem shall be microprocessors.

3.6.2.3.2 Performance. The dual processors of a switching center shall operate synchronously; one on-line and the other executing the same program on an instruction-by-instruction basis in a "ghosting mode." Only the commands of one processor shall activate and control the switching center at one time. The SCU shall determine which processor is on-line unless overridden by manual control. The processor shall be of sufficient capability to provide all the necessary system functions.

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3.6.2.3.3 Controls and indicators. The processor control panel for each processor shall provide access to all basic functions of the processor and shall be operable in the run, stop, on-line, standby, and off-line modes of operation. The controls and types by indicators for the processor control panel shall include, but not be limited to, control of the functions are listed below:

(a) Run mode - displays free running of processor.
(b) Displays of key registers.
(c) Compare stop - processor stops at end of instruction whose address is preset on manual panel switches.
(d) Single instruction step.
(e) Single machine cycle step.
(f) Register load and display controls.
(g) System reset.
(h) Parity error indications.
(i) Storage protect bit entry.
(j) Various input/output (I/O) device controls.
(k) Stop or attendant interrupt.
(l) Restart (from point of stopping).

3.6.2.4 Switchover control unit.

3.6.2.4.1 Description. The SCU shall provide the means to maintain synchronization between the two processors in a switching center and to automatically switch control of the center from a failed on-line processor to the off-line processor.

3.6.2.4.2 Performance. The SCU, upon detecting that a mismatch condition exists between the outputs of the two processors, shall initiate an on-line, self test routine within both processors. The results of these tests shall be decoded by the SCU. The processor with an error in the result shall be placed off-line and control of the center will be assigned to the operational processor. Should both processors pass the self test routine, the existing processor on-line status shall remain unchanged but an error message shall be generated indicating a mismatch. If a second consecutive mismatch occurs and both processors still pass the self-test routine, the processor on-line status shall continue unchanged, the SCU shall be turned off, and a unique error message shall be generated indicating this condition. The detection of error and switchover shall not interfere with calls in progress or the initiation of calls. The SCU shall also provide audible and visual indications of processor mismatch and synchronization failures. In addition, the SCU shall include the controls and circuitry required to synchronize the two processors when one or both are being placed back in service.

3.6.2.5 Memory.

3.6.2.5.1 Description. A semiconductor memory shall be associated with each processor. Each memory shall store program status information for use by the processor in performing the control subsystem functions specified in 3.6.2.2. The capacity of each memory shall be 32K words expandable to 64K words.

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3.6.2.5.2 Performance. An error detection and correction capability shall be incorporated in each memory for application to data transferred to and from the memory. The error detection and correction code provided shall be able to detect up to two errors and correct one error in any 24-bit word. Multiple errors in a word shall cause an alarm to be generated. Table VI lists the minimum acceptable requirements of the memory unit.

TABLE VI. Minimum acceptable memory unit requirements.

Characteristic	Requirement
Changeability and access speed	Adequate to accomplish the functions stipulated in this specification
Word length	30 bits (minimum) (with error detection and correction)
Modularity	16K words an increment
Size	32K words, expandable to 64K words
Power	35 watts maximum per 16K word memory increment.

3.6.2.6 Control programs. The control programs shall operate in the control subsystem and shall be responsive to, and provide for the control of, the IVCS subsystems. The programs shall be able to perform or interact with the hardware, which shall perform those functions required or implied as specified herein, including the following:

(a) Call processing.
(b) All terminal service features.
(c) Interface control.
(d) System start-up and recovery operations.
(e) Diagnostic and switchover operations.
(f) All required administrative and "housekeeping" operations.
(g) Man-machine features.

The control programs shall be organized on a functional basis consistent with the control subsystem architecture. The control programs or tables shall be capable of being modified or expanded to accommodate the configurations of the IVCS as specified in 3.3 and amenable to functional modification. Where the design includes programs that reside external to main memory, it shall be possible to load or reload such programs into main memory from the magnetic tape device (see 3.6.5.3). The control programs shall be provided in accordance with the data ordering document specified in the contract or order (see 6.2.2). The following paragraphs specify the minimum functions to be accomplished (see 6.2.1).

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3.6.2.6.1 Executive program. The executive program shall have overall responsibility for the control of each switching center and shall direct the execution of all program functions in order of their priorities. It shall perform such functions as program scheduling, control and sequencing, as well as elapsed-time monitoring. The executive program shall direct all functional programs and shall maintain control of all operations. The executive program shall include, but not be limited to, the following functions:

(a) Maintaining cognizance over the present status of all functional programs and determining function sequencing.
(b) Receiving, controlling, and scheduling the processing of all input/output (I/O) requests for the functional programs.
(c) Selecting, scheduling, and controlling the functional processing program modules, as needed, on a priority basis, from the highest level used in operations that can not be delayed to the lowest level used in non-time-dependent operations.
(d) Directly controlling the maintenance programs that detect processor, matrix, or other equipment failures and providing for automatic switchover to redundant equipment as needed.
(e) Exercising maintenance programs.

3.6.2.6.2 Maintenance programs. Maximum use shall be made of the control subsystem as a diagnostic device for the subsystems and peripherals of the IVCS. The maintenance programs shall provide the capability to isolate a fault down to a single printed wiring assembly or to a small group of replaceable printed wiring assemblies and shall be comprised of fault detection and fault isolation routines. It shall be permissible to store certain maintenance programs external to main memory, bringing them into main memory for execution only when required. Manual control of the maintenance programs shall be provided from the maintenance terminal (see 3.6.5.5). Table VII lists the maintenance terminal functions. Each maintenance program shall be identified by a 4-digit number as specified in 3.6.2.6.4.

TABLE VII. Maintenance terminal functions.

Item	Function
	The primary function of the maintenance terminal is to provide for manual control of the maintenance program. It is also used in a maintenance role to verify operation of specific portions of the system by placing or receiving calls.
1. Normal calls	The maintenance terminal in each switching center shall be able to initiate or receive any calls listed for a standard dial terminal.

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TABLE VII. Maintenance terminal functions. - Continued

Item	Function
2. Initiate on-line or off-line test routines a. Automatic processor testing b. Call net terminal d.c. test c. Switchover fault (on-line) d. Switchover fault (off-line) e. Tone generators f. Switchover comparator g. Receiver or sender h. Test connection and release i. Processor confidence j. Printer exercise k. Memory comparison l. No skip messages m. Cancel	Allows operator to select test routine by keying a 4-digit number and adding additional control parameters as required. (Examples of test routines)
3. Permit changes: a. Shore b. Radio c. Conference d. Override e. General announcing f. Call forwarding g. Abbreviated addressing (AA) h. Alternate number i. Restriction category	Allows for the changing of a permit by specific coded input to add or delete a permit.
4. Data gathering routines (off-line)	Allows operator to initiate data gathering for a switching center or selected terminals and designate the interval and period for printout. A period ranges from 1 to 99 hours. An interval ranges from 1 to 99 minutes. Period information is a summary of interval data. Interval data printing can be suppressed.

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TABLE VII. Maintenance terminal functions. - Continued

Item	Function
a. General data gathering	This information is collected for a switching center. Interval data: 1. Total calls originated since start of data gathering 2. Total calls originated/interval 3. Total conference calls originated/interval 4. Receiver/sender queue count an interval 5. Total trunks busy/interval 6. Time of each interval. Period data: 1. Total calls originated, conference calls, queued calls, and trunks busy period 2. Average number calls originated, conference calls, queued calls, and trunks busy interval in that period 3. Peak number calls originated, conference calls, queued calls, and trunks busy interval in that period and the time of that peak.
b. Discrete data gathering	This information is collected for specified terminals. Interval data: 1. Total calls originated interval 2. Total calls received interval 3. Total times busy interval 4. Total net calls interval 5. Time of each interval. Period data: 1. Total calls originated, calls received, times busy, and net calls per period 2. Average number calls originated, calls received, times busy, and net calls per interval during that period 3. Peak number calls originated, calls received, times busy, and net calls per interval during the period and the time of the peak 4. Average duration of all calls originated and received during the period 5. Duration of the longest call made during the period followed by the time of termination.

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TABLE VII. Maintenance terminal functions. - Continued

Item	Function
5. Memory dumps (print-outs)	Allows routine printouts of areas in memory bank 1 or 2 where the area specified is between the two addresses dialed in from the Maintenance Terminal.
6. Read memory	Ability to read, at the maintenance command, any word in memory bank 1 or 2.
7. Write memory	Allows changing of any word in working storage and any word in the permanent tables when the MEMORY WRITE switch is in the ON position.
8. Emergency report changes	Allows changing emergency reporting station whenever the Readiness condition of the ship is to be changed.
9. Equipment status	Allows operator to request a printout of certain equipment status (out of service or return to service).
10. Subscriber number change	Allows operator to add or delete a subscriber or to change the number of a subscriber.

3.6.2.6.3 Main traffic program. The main traffic program shall include all routines and subroutines specifically prepared to accomplish the switching and communications functions of the IVCS. These program units shall be comprised of the routines necessary to perform the input (terminal service requests), output, and internal processing functions and the management of associated dynamic tables. Changes to tables assigning subscriber features, system accesses, and restrictions to individual terminals shall be possible through the maintenance terminal (see 3.6.5.5). Table VII lists the types of table or permit changes possible through use of the maintenance terminal. The main traffic program shall include the call processing and signaling and supervision routines as specified in 3.6.3.2 and provisions and features listed below.

3.6.2.6.3.1 Progressive conference. It shall be possible to assign the progressive conference capability to any dial terminal. This type of conference call shall be set up as follows: the calling party (conference originator) will dial a special conference code (9703) after which a dial tone shall be received. The connection between the calling party and the first called party (first conferee) shall occur as the result of dialing and system response identical to a regular 2-party call. To progressively expand the conference, the calling party will depress the OFF button followed by an immediate depression (within 1 second) of the ON button (equivalent to the depression and release of a "hook switch"). Recall dial signal shall be received and the calling party

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will dial the second party to the conference. The calling party shall receive a ringback tone, busy tone, or unavailable tone. When ringback is received the calling party can return to the conference by once again flashing the ON/OFF buttons. The called party will be connected to the conference if the phone is answered within a 10- to 15-second ring period. If the called party answers before the calling party returns to the conference, the two parties can converse and then when the calling party flashes the ON/OFF buttons, both parties will be connected to the conference. If the called conferee is busy or unavailable, the proper tone will be returned to the conference originator for 3 seconds and then he will be reconnected to the conference. The calling party shall then be able to remain in the conference connection or repeat the ON/OFF buttons cycle and obtain the recall dial signal to further expand the conference. The following additional requirements shall be met in providing the progressive conference service:

(a) Any party may disconnect from the conference without disturbing the rest of the conference. Therefore, all parties shall manually disconnect. When the last party disconnects, the conference is broken down.
(b) A full-duplex transmission arrangement shall be provided, for example, if one party is speaking, another party may speak simultaneously by also pressing his push-to-talk (PTT) switch.
(c) Up to 10 six-party random add-on or progressive conferences may be in progress in each switching center at one time. A conference initiated in one center may bring conferees in from the other center by trunks. Up to five random add-on or progressive conference calls may be in the process of setup in each center at any one time.
(d) If a conferee has his handsfree loudspeaker in the handsfree mode, he will not hear another conferee who has his handsfree loudspeaker in the handsfree mode. He will only hear conferees who activate their PTT switches.

3.6.2.6.3.2 Call transfer. The call transfer capability shall be available to all subscribers who are involved in a 2-party call. Subscribers currently connected to nets, involved in conferences, or being serviced by the attendant will not have this capability. The call transfer will be initiated by a subscriber flashing the ON/OFF buttons. The system shall return the recall dial tone and the call transfer initiator may dial a new party. If the new party is out-of-service, or busy on another call, supervisory tone will be returned to the call transfer initiator for 3 seconds. The call transfer initiator must flash the ON/OFF buttons to return to the original party. Privacy override cannot be initiated as part of the call transfer function. Calls cannot be transferred directly to shore parties. Call transfer to multiple party calls is not allowed. A maximum of three subscribers can be connected at any one time through the call transfer capability. If the new party is idle, he will be rung and a ringback tone will be sent to the call transfer initiator. If the call transfer initiator disconnects, the new party and the original party (other end of call transfer initiator) will be connected as a normal, two party call. If the new party answers within 10 to 15 seconds, the call transfer initiator and the new party will be connected. The call transfer initiator can then connect

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the other original call member by flashing his ON/OFF buttons. If the new party does not answer in that time the call to the new party will be dropped. The call transfer initiator can then connect the original call member by flashing his ON/OFF buttons. If the new party answers, the call transfer initiator may elect to consult with the new party only. If the new party then disconnects, the call transfer initiator must flash his ON/OFF buttons to return to the other original call member. Release by any party in call transfer will not disturb the remaining parties. The call transfer function gives the following capabilities:

(a) Transference of a call to another position.
(b) Consultation with a third party while keeping the original party on hold.
(c) Three-party calls.

3.6.2.6.3.3 Preset conferences. The preset conference is a special feature which is provided to automatic terminals and associated loudspeakers to expedite the transmission of urgent tactical information. Up to 14 preset conferences (one at a time) may be implemented from any automatic terminal through abbreviated address dialing. Privacy override is inherent to the preset conference feature. Any one preset conference may include up to six called terminals. The called terminals will be other automatic terminals only. When a conferee has his loudspeaker in the handsfree mode, he may only hear those conferees who activate their PTT switches, not another conferee with his loudspeaker in the handsfree mode. When a preset conference originator hangs up, the conference is broken down and all parties are disconnected from the conference and restored to original connections; however, any other party may hang up without disturbing the conference.

3.6.2.6.3.4 Access restriction. It shall be possible to divide all IVCS lines into access restriction categories for the purpose of controlling intercommunications between functional subscriber groups. Provisions shall be made for up to five categories and any line may be assigned to any of the categories. A line assigned to a specific category shall have unrestricted access to other lines in that category and access to other designated categories. The specific relationship between categories for each system configuration will be specified (see 6.2.1). An example of the relationship between categories follows:

Restriction category	Can call
A	A C D E
B	B
C	A C E
D	A D E
*E	A C D E

* Category E is assigned to line load control.

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3.6.2.6.3.5 Line load control. During periods of traffic exceeding a predetermined level, subscribers in certain restriction categories will be barred from originating trunk calls, although they shall always be able to receive calls and place emergency calls. The program will keep a gross count of all trunk calls in progress in order to automatically determine when such traffic conditions exist. When traffic returns below another preset level, the program will restore normal service to these subscribers.

3.6.2.6.3.6 Privacy override. It shall be possible to assign the override capability to any dial terminal. A subscriber using a terminal which has been assigned the privacy override capability may initiate an override call in two ways. After receiving a dial tone the subscriber may depress the override button (OV) which will send a special prefix code indicating an override call and then key in the called party's address; or, after keying the called party's address and receiving a busy tone, he may depress the OV button within 4 seconds of keying the last address digit. The rest of the call shall proceed as follows, depending on conditions of the switching center and the called party. (Override privileges do not permit a user to cancel any assigned access restrictions.) (Table VIII summarizes the system response to dial terminal signaling under various connected call conditions.)

(a) Common equipment busy. If the override call cannot be processed because of this condition, then a busy tone shall be returned to the calling party.
(b) Called party idle. In this case the call shall be handled as a normal call.
(c) Called party busy on a line-to-line call. In this case the control program shall establish an override connection with the desired party, in which the overriding party is connected in parallel with the subscriber presently connected to the overridden party. (NOTE: The overriding party is defined as the calling subscriber; and the overridden party is the subscriber being called.) A 1-second override tone, placed on the existing connection, shall precede override action. Once the connection has been established, release by any party will not disturb the remaining connection.
(d) Called party busy on a net call. In this type of override, a connection shall be established, with override tone, from the overriding party to the overridden party and all the members of the net. The overridden party will not be disconnected from the net.
(e) Called party busy on a progressive conference call. This type override will be similar to a net call override (see 3.6.2.6.6(a)).
(f) Called party busy on general announcing, shore calls, ship's radio calls, or preset conference. In these cases, override will not be possible. The calling party shall receive a busy tone if he attempts to override.
(g) Called party involved in override call. A user cannot override a call already being overridden. The calling party shall receive a busy tone. Conference calls can be overridden until a maximum of seven conferees are reached.

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(h) In no case will any conference call, preset or progressive, be allowed to override another conference call.

(i) In all cases, when the overriding party terminates his call the overridden connection shall not be disturbed.

(j) Calls originated from automatic terminals. In this case, all of the above conditions, (a) through (j), shall occur; however, privacy override shall be accomplished automatically when abbreviated dialing is used, without the necessity of depressing the override pushbutton.

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TABLE VIII. System response to dial terminal signaling.

Dial terminal	Connected call conditions												
Method of dialing	Idle	On station call	On station w/ovr	In confer (random) (progressive)	In pre-set confer	Any confer w/ovr	In net	In net by ovr	On radio	On shore line	Using announcing	Using IADS	Out of service
4 digit	C	A	A	A	A	A	A	A	A	A	A	A	A/U
With privacy override	C	P	A	P	A	P*	P	A	A	A	A	A	A/U
AA													
Standard dial	C	A	A	A	A	A	A	A	A	A	A	A	A/U
Automatic	C	P	A	A	A	A	P	A	A	A	A	A	A/U
Progressive conference	C	A	A	A	A	A	A	A	A	A	A	A	A/U

Action key:

A - Route to alternate phone if assigned, otherwise busy tone returned
C - Connect dialed call
N - Not connected and no indication
P - Connect with privacy override tone
U - Unavailable tone returned
P* - Up to maximum number of conferees plus one

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3.6.2.6.3.7 Call forwarding. Dial terminals assigned this capability shall be able to have all incoming calls routed to another dial terminal through program control and subscriber implementation. Automatic terminals cannot call forward but calls can be forwarded to automatic terminals. To implement call forwarding, the subscriber shall be required to dial a special service code followed by the 4-digit number of the terminal to which the calls are to be forwarded. To return to normal service, the subscriber must again dial the same special service code followed by his own number (home). Subscriber action shall be conducted from the home terminal. The implementation of call forwarding shall neither affect any operation of the terminal to which calls are forwarded nor affect outgoing calls from the home terminal.

3.6.2.6.3.8 Assigned access. It shall be possible to assign selected dial terminals access to any combination of the following: individual nets, man-on-the-move (MOM) channels, general announcing systems, ship's radio, and one of four shore line modes. Terminals assigned such accesses shall be able to obtain the desired connection by keying the appropriate number as identified in the numbering plan (see 3.6.2.6.4). Terminals which attempt to complete a call to a connection to which access has not been assigned shall receive an unavailable tone. Specific assignment of terminal accesses shall be specified (see 6.2.1).

3.6.2.6.3.9 Busy party alternate. It shall be possible to designate up to three alternate terminals (numbers) for selected dial terminals. The total number selected will be up to 20 percent of the switchable terminations or as specified (see 6.2.1). In the event the primary terminal is busy when called, the alternate terminals shall be checked in order and the first idle alternate rung. Local alternates will be checked first followed by remote alternates. If all alternates are busy, the calling party shall get a busy tone back. If the calling party then presses OV (after getting a busy tone), the last terminal in the alternate series shall be overridden. If the calling party initiates a call with OV, the call shall only go to the called party and not to any alternates.

3.6.2.6.3.10 Abbreviated addressing. The capability of abbreviated addressing shall be programmable to any individual dial terminal. The 12- or 16- pushbutton keyboard will be used for all dialing. The abbreviated addressing key (A) shall be depressed to initiate each abbreviated addressing (single-digit) call.

3.6.2.6.3.11 Call director mode. The call director mode of the multiline terminal and its associated terminals shall function as follows.

(a) The call director operator shall occupy one unique position of the multiline terminal and shall be capable of answering four other terminals. The other four terminals shall be in parallel with the four appearances in the associated multiline terminal.
(b) When the operator is present, calls to any of the four associated terminals shall be routed to the operator position on the multiline terminal. The operator shall be able to answer, screen, and then forward calls to the called party, as required, by utilizing the call transfer function.

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(c) If the operator is busy, a call to one of the other parties shall be sent directly to that party. If that party is not there or does not answer, the operator still has the option to answer the call by depressing the proper flashing button on the multiline terminal.

(d) The operator shall be provided with a 4-digit dial up code to alert the control program that the operator's position is to be vacant. The software shall then direct all multiline terminal incoming calls to the other parties directly. The operator shall also be provided with another 4-digit code to alert the software that the multiline terminal is to be manned and to route incoming calls to it.

3.6.2.6.4 Address numbering plan. The address numbering plan consists of a universal, closed numbering plan in which each terminal (dial and net), net, interface channel, service code, or maintenance code is identified by a discrete 4-digit number. The address numbers are used in the main traffic program for identification purposes and by the subscriber for service requests. The numbering plan is divided into two parts: a fixed or reserved set of numbers and a directory set of numbers (see 6.2.1).

3.6.2.6.4.1 Reserved numbers. This set of numbers will be common to all ship classes using the IVCS.

Number	Function
0	Shore line control terminal (SLCT)
2211	Emergency reporting number
2212	Reserved
2213	Reserved
2214	Reserved
84XX	Reserved for interface with other ships systems
87RR	Radio with number RR where R = 1-9
88NM	Net or MOM with number NM where N = 1-9 and M = 0-9 (emergency net numbers excluded)
8891	Net number of emergency reporting net
8892	Net number of emergency reporting net
8893	Net number of emergency reporting net
8894	Net number of emergency reporting net
9007	Ringdown switchboard request
9009	Private automatic branch exchange (PABX) line request
910X	Ringdown instrument request where X = 1 to 9
96PP	PA request with number PP where P = 0-9
9700	Call forward request/cancel
9703	Progressive conference request
99MM	Maintenance request from maintenance terminal with number MM where M = 0-9

3.6.2.6.4.2 Telephone directory numbering. The telephone numbers assigned to terminals may be any number which is not a reserved number. The assignment of numbers to both dial and call net terminals will be based on the following plan:

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(a) First number significance. The first number will indicate ship department.

1 - Executive department
2 - Operation department
3 - Navigation department
4 - Supply department
5 - Combat system department
6 - Engineering department
7 - Communications department
8 - Medical/dental department

(b) Second number significance. The second number will indicate the division within the ship department.
(c) Third number significance. The third number will indicate a common function or common space.
(d) Fourth number significance. The fourth number will indicate the switching center, when there is a dual center configuration, to which a particular terminal is connected. An odd fourth digit will indicate the terminal is connected to the forward center and an even number fourth digit will indicate the terminal is connected to the after center. This also identifies a discrete phone assigned to a space or common function. The odd-even fourth digit designation does not pertain to single center installations.

3.6.2.6.4.3 Jackbox terminal numbering. Since jackbox terminals cannot be called, they shall be identified by a one alpha and three numeric code designation vice a telephone number. The fourth digit will identify the IVCS switching center to which the jackbox terminal is connected (odd for the forward center and even for the after center). The alphanumeric identifiers assigned will be sequential according to termination number and center. As specified in 3.6.2.6.4.2(d), this pertains only to dual center configurations.

3.6.3 Switching subsystem.

3.6.3.1 Description. The switching subsystem, under the control of the control subsystem, shall complete the audio paths between terminals and from terminals to other ship systems by the distribution subsystem. It shall also provide all required signaling and supervision to terminals, trunks, SLCTs, and interface units as received from the control subsystem. The subsystem shall consist of termination units for matrix appearances and switching matrices.

3.6.3.2 Performance. The switching subsystem shall provide for nonblocking traffic flow among all matrix appearances in each switching matrix assembly.

3.6.3.2.1 Signaling and supervision. The switching subsystem shall provide the path for signals transmitted from terminals to the control subsystem and from the control subsystem to terminals. These signals are identified in tables IX and X.

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TABLE IX. Signaling and supervision from terminal to control subsystem.

Signal	Function	Description
Service request		
Dial terminal	Indicates need for service	Change in voltage on supervisory line from 13 Vd.c. to 5 Vd.c.
Call net terminal	Indicates need for service	11-bit digital signal at 2400 bits per second consisting of: start, 8 data bits, parity, and stop
Addressing		
Dial terminal[1/]	Identifies service desired	DTMF signals Digit — Tone frequency (Hz) 1 — 697 and 1209 2 — 697 and 1336 3 — 697 and 1477 4 — 770 and 1209 5 — 770 and 1336 6 — 770 and 1477 7 — 852 and 1209 8 — 852 and 1336 9 — 852 and 1477 0 — 941 and 1336 OV — 941 and 1477 A — 941 and 1209 W — 697 and 1633 X — 770 and 1633 Y — 852 and 1633 Z — 941 and 1633
Call net terminal	Identifies called party	11-bit digital signal
Release request		
Dial terminal	Indicates end of call	Change in voltage level on supervisory line from 5 Vd.c. to 13 Vd.c.
Call net terminal	Indicates end of call	11-bit digital signal

1/ A calling party shall hear DTMF signals being forwarded by a trunk for calls to called parties connected to remote switching centers in dual center configurations.

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TABLE X. Signaling and supervision from control subsystem to terminals.

Signal	Function	Description
Dial tone		
Dial terminal	Indicates availability of a R/S to process call	6-second, 500-Hz tone
Call net terminal	Not applicable	Not applicable
Ring forward signal		
Dial terminal	Alerts called party of incoming call	Signal alternating between floating (open circuit) and ground at a 2.5-second, on-off interval used to activate ring circuitry and illuminates activity indicator on the terminal
Call net terminal	Alerts called party of incoming call	11-bit digital signal used to activate ring circuitry and to illuminate activity indicator on the terminal at the ringback rate
Net status signals		
Dial terminal	Not applicable	Not applicable
Call net terminal	Indicates status of terminals associated with call net buttons	11-bit digital signal to light appropriate call net buttons
Ringback signal		
Dial terminal	Indicates progress of call to the calling party	2500-Hz tone at 2.5-second, on-off rate similar to but not synchronous with the ring forward signal
Call net terminal	Indicates progress of call to the calling party	On-off flashing of selected call button at ringback rate (2.5-second, on-off)

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TABLE X. Signaling and supervision from control subsystem to terminals. - Continued

Signal	Function	Description
Busy signal		
Dial terminal	Indicates to the calling party that the called party is on another call	500-Hz tone alternating 1/4-second off and 1/4-second on for a 3-second duration
Call net terminal	Indicates to the calling party that the called dial terminal is busy on a call other than the assigned net. Busy signal is not applicable to calls placed between call net terminals	On-off flashing of selected call button at busy signal rate (1/2 second on and 1/2 second off)
Recall signal		
Dial terminal	Indicates calling party has regained R/S for further signaling after flashing the ON/OFF buttons	A burst of three 1/4-second on-off 500-Hz tone signals followed by dial tone
Call net terminal	Not applicable	Not applicable
Unavailable signal		
Dial terminal	Indicates that the calling party has dialed a number for which the calling terminal does not have access permit or that the called terminal is out of service	Tones of 500- and 1100-Hz alternating at 1/4-second intervals for a duration of 3 seconds

3.6.3.2.2 Subscriber calling. It shall be possible for subscribers to initiate service and obtain the response described below:

3.6.3.2.2.1 Dial terminal, line-to-line calls. The calling party requests service by depressing the ON button, activating the terminal. A dial tone shall indicate that the switching subsystem is ready to accept dialed digits. The calling party indicates the desired service by keying the terminal. During dialing, the dial tone is turned off (indicating that dialing is being accepted) after the first digit is received. After dialing is completed, the control subsystem shall process the digits and shall then attempt to establish a connection between the calling and called parties. In the case of parallel dial terminals, a ring inhibit line will be connected between the terminals to prevent ringing of a parallel terminal when the active terminal is originating

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a call or using PTT. If the called party is not busy, a ringback tone shall indicate that the called party is being rung. If the called party or the common equipment is busy, a busy tone shall be sent to the calling terminal. In the case of an automatic terminal calling an automatic terminal through abbreviated dialing, if the called terminal is busy, automatic privacy override will be initiated as if the calling automatic terminal had initiated override. If the called terminal is out of service or the call is not permitted, an unavailable tone shall be returned to the calling terminal. The called terminal, if not busy, shall be rung until: the call is answered, the calling party terminates the call, or after a time-out of 12 to 15 seconds has occurred, whichever comes first. In the case of an automatic terminal, there will be one ring after which the connection is completed. On established calls, if override was requested and authorized a 1-second burst of override tone shall indicate that privacy override is being initiated and the conversation is about to be entered by the overriding party.

3.6.3.2.2 Dial terminal line-to-net calls. The calling party activates the terminal, receives a dial tone, and keys the terminal for the desired service. If the dialed number represents a net, the calling party shall be connected to the net. If the dialed number represents a call net terminal, the calling party shall be immediately connected to the net, and if idle, the called terminal shall be rung until the call is answered or a time-out of 12 to 15 seconds has occurred, whichever comes first. When the call net terminal answers, it shall be immediately connected to the net. If the calling party does not have access to the net, or the called terminal is out of service, an unavailable tone shall be returned to the calling terminal. If the call net terminal is already busy on the net, communications can commence.

3.6.3.2.2.3 Call net terminal, net calls. Call net terminals are activated by either depressing the ON button or one of the call net buttons. Depressing the ON button shall immediately connect the terminal to the preassigned net. Depressing one of the terminal buttons assigned to another call net terminal shall immediately connect the terminal to the preassigned net and cause the corresponding call net terminal to be rung. In all cases, whenever a call net terminal is connected to the assigned net, the call buttons will be lit for all assigned terminals that are already connected to that net. When a call net terminal is called to the net and the calling party has a call button assignment on the called call net terminal, that button will be flashed at the ring rate to indicate the identity of the calling party.

3.6.3.2.2.4 Call net terminal, line calls. It shall be possible for call net terminals to initiate calls to specific dial terminals by preassigning the address of the dial terminal to the call net terminal button. Depression of a call net button so assigned shall cause the call net terminal to be activated and connected to the net, and the corresponding dial terminal to be rung, if idle, for a standard 12- to 15- second period. If the dial terminal is answered, it will be connected to the net. If the dial terminal is busy, the corresponding call net button shall be flashed at the busy rate.

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3.6.3.2.3 Emergency reporting calls.

3.6.3.2.3.1 Emergency reporting nets. A capacity for four emergency reporting nets shall be provided to receive emergency calls from any dial or automatic terminal. Each net implemented shall be identified by an emergency telephone number in addition to the net number. Emergency numbers and net numbers, with numbers reserved for emergency, are as follows:

Dial emergency number	Net dial number
2211	8891
2212 (Reserved)	8892 (Reserved)
2213 (Reserved)	8893 (Reserved)
2214 (Reserved)	8894 (Reserved)

3.6.3.2.3.2 Responsible dial terminal. One dial terminal shall be assigned to each emergency net (responsible dial terminal) and be responsible for handling incoming emergency calls. When any dial or automatic terminal dials the emergency number, the responsible dial terminal shall be rung. When this terminal is activated, the calling terminal and the responsible dial terminal shall be automatically connected to the emergency net corresponding to the emergency number dialed. If the responsible dial terminal is already connected to the emergency net, the calling terminal is automatically connected to the emergency net. An additional dial terminal also dialing the emergency number shall be automatically connected to the net. The responsible dial terminal shall then respond to both calls over the emergency net. If the responsible dial terminal is busy on a call to some location other than an emergency net, the responsible dial terminal shall be alerted that there is an emergency call by receiving the override tone (continuous 1100 Hz) and override will be executed connecting all three parties to the emergency net.

3.6.3.2.3.3 Call termination. When the originating (calling) terminal terminates an emergency call by deactivating the terminal, its connection to the net shall be released. The responsible dial terminal connection to the emergency net shall be maintained. When the responsible dial terminal deactivates, its connection to the net shall be broken.

3.6.3.2.3.4 Designation of responsible dial terminals. The specific location of the responsible dial terminal depends on whether the ship is at sea or in port and the manner in which it is moored in port. It shall be possible to change the designated location of the responsible dial terminal using the maintenance terminal.

3.6.3.2.3.5 Monitor stations. It shall be possible to connect one or more net terminals equipped with a loudspeaker to the emergency reporting net to continually monitor the net. It shall be possible to handle emergency calls at any of the monitor stations if necessary.

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3.6.3.2.3.6 Net terminal access to emergency reporting nets. Net terminals shall have access to emergency reporting nets when so assigned and wired. Emergency calls originated from net terminals shall cause the terminal to be connected to the emergency net but shall not cause the responsible dial terminal to be connected to the net. Emergency calls from net terminals shall be handled by one of the monitor stations which will be active on each emergency net at all times.

3.6.3.3 Switching matrices.

3.6.3.3.1 Description. The switching matrices shall provide the means to accomplish the specified switching actions under the control of the control subsystem. The switching actions shall include the routing of voice signals between any termination unit and any other termination unit associated with the switching matrix.

3.6.3.3.2 Performance. Each matrix shall be a nonblocking time-division matrix and shall be designed to provide full-duplex operation and conferencing. Each matrix shall be configured so that failures incrementally reduce service as opposed to producing catastrophic failure. The matrices and line circuit terminations units, under control of the on-line processor, shall also provide for automatic line circuit idle or loop-back test of each line.

3.6.3.4 Termination units.

3.6.3.4.1 Description. A termination unit is that circuitry which allows a matrix appearance to be used in conjunction with a line, net, trunk, or interface connection. The termination unit shall include the circuitry for impedance matching, filtering, and analog-to-digital (A/D) and digital-to-analog (D/A) conversion necessary to interface the matrix appearance with the associated line or trunk. Multiple termination units shall be mounted on a single printed wiring assembly (PWA) where practical.

3.6.3.4.2 Performance. Each termination unit shall contain filtering, A/D, and D/A circuitry which will pass voice signals through the switching center in conformance with the transmission criteria of table I. The output of each termination unit shall provide a 0 dBm signal to the connected terminal, net, trunk, or interface. The PTT signal shall be provided as an additional parallel signal.

3.6.4 Distribution subsystem.

3.6.4.1 Description. This subsystem shall consist of the MDF and inter-rack cabling. The distribution subsystem shall distribute operating power to terminals and SLCT. In addition, the distribution subsystem shall provide the physical connections from terminals and ship systems to the switching subsystem. This subsystem shall also be the termination point for inter-IVCS switching center when more than one center is required.

3.6.4.2 Performance. The distribution subsystem shall provide the required electrical connections to support given the terminal capacity and traffic-handling requirements specified in 3.5.2.

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3.6.4.3 Main distribution frame.

3.6.4.3.1 Description. The MDF shall accommodate all IVCS switching center terminal, net, interface, and inter-IVCS switching center terminations.

3.6.4.3.2 Performance. The MDF shall be of modular design. Each powered termination shall be individually overload and short circuit protected. The MDF shall provide access to each termination. The MDF will include the electrical and physical interfaces for the ship cabling and the connector field for the interrack cabling. These interfaces shall not degrade the transmission requirements specified in 3.5.2.4. The MDF field connectors shall be in accordance with MIL-C-5015, MIL-C-10544, and MIL-C-28840, as applicable. The MDF shall be designed to provide electromagnetic interference (EMI) integrity. The design shall provide for ship cable to enter the MDF through the bottom of the enclosure and allow for the twisted pair shields to be stripped and grounded within the confines of the enclosure. The twisted pairs shall be terminated on terminal boards which shall be in accordance with MIL-T-55164 and MIL-STD-1277, as applicable. Assignment of terminal board locations for the 19 twisted pairs of 2SWU-19 cable shall be in contiguous vertical locations where possible in accordance with MIL-C-915/49. Provisions shall also be made to terminate spare wires contained in each cable but not necessarily in the same contiguous location. Provisions shall be available to terminate 54 cables with an expansion capability to 72 cables.

3.6.4.4 Interrack cabling.

3.6.4.4.1 Description. The interrack cabling shall provide all power and electrical connections which interconnect manufacturer-supplied elements of the switching center. Additionally, the interrack cabling shall include the power, audio, and signaling cabling from the center(s) to the SLCT.

3.6.4.4.2 Performance. The cabling and connectors provided shall not degrade the transmission requirements specified in 3.5.2.4. Interrack cables, wires, and connectors shall be in accordance with MIL-C-915, MIL-C-28840, MIL-W-81044, MIL-W-16878 type B, MIL-STD-454, MIL-C-5015, and MIL-C-10544, as applicable.

3.6.4.5 Net switching capability. The capability shall be provided to interconnect any net to any other net. The interconnection of nets shall be under the control of the operator. A visual indication shall be provided which displays the nets which are interconnected.

3.6.5 Peripherals.

3.6.5.1 Description. The control subsystem shall interface with a magnetic tape cartridge unit, line printer, and maintenance terminal.

3.6.5.2 Performance. The performance characteristics for these peripherals are identified in 3.6.5.3 through 3.6.5.5.

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3.6.5.3 Magnetic tape cartridge unit. A four-track (selectable) cassette-type magnetic tape cartridge unit with the necessary supporting logic shall be provided for loading the control programs identified in 3.6.2.6 and recording outputs from the control subsystem. The tape unit shall be capable of operation either manually from front panel controls or automatically from signals generated by the processor. The magnetic tape cartridge shall be capable of holding a minimum of 20 million bits continuous recording. The operating speed shall be 30 inch-second, write or read, and 90 inch-second, rewind or search. Manual front panel controls shall be provided to assist in fault isolation and for operator convenience in rewinding tape. The interface between the processor and the tape unit shall be EIA-RS232 (see 6.2.1).

3.6.5.4 Line printer. A printer with necessary supporting logic shall be provided for maintaining a permanent record of transactions including operator-generated instructions, the processor response to instructions, processor-generated action printouts, and any other data directed to it by the operator. The printer shall be capable of handling standard stock issue 80-column fan-fold, tractor feed, single- to four-ply paper. The printer shall have a printing speed of at least 300 lines a minute (based on a minimum of an 80-character line). Full-sized characters shall be 0.100 $\pm$ 0.005 inches high by 0.075 $\pm$ 0.002 inches wide and spaced 10 an inch horizontally. Line spacing shall be 6 an inch. The printer shall be capable of printing the ASCII standard 64-character graphic subset (upper case). Operator maintenance aids shall include visual indicators for conditions such as no paper, low paper, and paper gate open. The interface between the processor and line printer shall be EIA-RS232.

3.6.5.5 Maintenance terminal. The maintenance terminal shall be a standard dial terminal capable of performing the functions listed in table VIII and 3.6.2.6.2. The capability shall also be provided to use an AN/USQ-69(V) data terminal set with an EIA-RS232 interface for the maintenance functions in lieu of the standard dial terminal. The selection of maintenance terminal and, if selected, the detailed description of the AN/USQ-69(V) shall be provided (see 6.2.1).

3.6.6 Terminals and accessories.

3.6.6.1 Description. The types of terminals that shall be utilized in the IVCS are identified in table XI.

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TABLE XI. Terminal types.

Type	Class	Scope of function
I. Dial	A. Standard 12-button	Single line terminal for administrative and tactical use.
	B. Standard 16-button	Same as A above but with 16 versus 12 pushbuttons.
	C. Automatic 12-button	Single line terminal with automatic operation (auto turn on or turn off from a remote terminal) with an integral or separate handsfree loudspeaker. Intended for rapid point-to-point and preset conference operation.
	D. Automatic 16-button	Same as C above but with 16 versus 12 pushbuttons.
	E. Two-line	Two-line terminal with access to two separate IVCS appearances. For use by one operator for simultaneous, independent two-channel operation.
	F. Multiline	A standard class B terminal and a multiline unit in a common enclosure. For use by one operator in one of two modes: a. Selective line monitoring and selective talk. b. Secretarial function in a multiline office.
II. Net	A. Call net 4-button	Single line, dedicated net terminal with a call capability of up to four other call net or dial terminals on the same net.
	B. Call net 8-button	Same as A above but capability to call eight other call net or dial terminals on the same net.
	C. Call net 16-button	Same as A above but capability to call 16 other call net or dial terminals on the same net.
	D. Jackbox	Single net terminal for dedicated net operation.
III. SLCT		Shore line control terminal, a terminal for use by the attendant for processing calls in the manual mode between IVCS subscribers and shore-based subscribers.

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3.6.6.2 Performance. The detailed performance characteristics and interface characteristics for terminals and accessories shall be provided (see 6.2.1).

3.6.6.3 Dial terminal. A dial terminal shall have the capability of obtaining direct dial service to other dial terminals, automatic terminals, and call net terminals. In addition, direct dial service will be available for access to nets, shore lines, and other ship's systems interfacing with the IVCS. Specific IVCS special features available for use by a dial terminal will be limited by the physical configuration of the terminal table XI and the accesses or permits made available to that terminal through the control programs within the control subsystem as specified in 3.6.2.6.3.

3.6.6.4 Net terminal. Net terminals will be used to connect subscribers having a common task or interest. These terminals will be connected to a circuit which permits all terminal users to talk and listen to all other terminals which have been similarly connected.

3.6.6.5 Accessories. Accessories available for use on an individual terminal are listed in table XII (see 6.2.1).

TABLE XII. Accessory to terminal interface (see 6.2.1).

Accessory	Type I						Type II			
Class	A	B	C	D	E	F	A	B	C	D
Handset, H319/U	X	X	X	X	X	X	X	X	X	X
Headset, H320/U	X	X	X	X	X	X	X	X	X	X
Headset, (heavy)	X	X	X	X	X	X	X	X	X	X
Headset, (two-channel)					X					
Headset, (single ear)	X	X	X	X		X	X	X	X	X
Intercommunication station, LS-613/STC-1 (standard loudspeaker extension)	X	X			X	X	X	X	X	
Intercommunication station, LS-651/STC-1 (loudspeaker extension, handsfree) or LS-614B/STC-1	X	X	X	X	X	X	X	X	X	
Footswitch	X	X	X	X	X	X	X	X	X	X

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3.6.7 Shore line interface.

3.6.7.1 Description. The shore line interface shall provide telephone line connectivity between the ship and shore telephone exchanges. The total number of interface lines will be specified (see 6.2.1). One interface electronics module shall be provided for each shore line. The SLCT will be considered a part of the shore line interface. Each system configuration shall be registered in accordance with Part 68 of the Federal Communications Commission (FCC) Rules and Regulations (Telephone Equipment Registration Program).

3.6.7.2 Performance. The shore line interface shall provide connections between the IVCS and shore lines of the following types:

(a) Four-wire AUTOVON direct exchange (DX) lines (total six wires including E&M leads)
(b) Two-wire standard commercial loop start lines
(c) Lines from military switchboards or terminals of the ringdown type.

It shall be possible to designate (mark) any line or group of lines for only incoming or outgoing calls or for both. The interface shall provide four modes of operation.

3.6.7.2.1 Manual. Under the manual mode a subscriber shall dial "0" to reach the attendant at the SLCT. The attendant answers the call, determines the shore number desired, and places the call to the shore. After the two parties are connected by the switching subsystem, the attendant shall depress the release pushbutton which shall free the previously utilized SLCT matrix appearance for subsequent calls. Incoming calls from the shore will be connected in the same manner. This mode of operation can be utilized with all types of shore telephone lines.

3.6.7.2.2 Private automatic branch exchange (PABX). Under the PABX mode, a group of lines from among the shore interface lines will be designated as PABX lines. This group shall appear to the switching center as a line hunt group for outgoing calls. A subscriber with the required software permit will be able to dial direct to a central office after dialing the required access code. After dialing central office access code, the subscriber shall receive a second dial tone after which the desired number may be dialed. For incoming calls, the PABX designated lines shall be handled in the same manner as for the manual mode. The PABX mode can be utilized with two-wire standard commercial loop shore telephone lines only.

3.6.7.2.3 Direct inward dialing. A group of shore interface lines can be designated for inward dialing. The purpose of inward dialing is to allow a shorebased subscriber to access the switching subsystem. Inward dialing requires that a separate number be assigned to the line at the shore telephone exchange. When this number is dialed the subscriber shall receive a dial tone from the ship which will allow the caller to dial the subscriber number. Inward dialing requires DTMF signaling from the shore subscriber to the IVCS switch. Inward dialing can be utilized with four-wire AUTOVON and two-wire standard commercial loop shore telephone lines only.

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3.6.7.2.4 Dedicated lines. Any shore interface line can be utilized as a dedicated line. When a shore interface line is designated as a dedicated line it will be identified as a shipboard-dedicated line within the shore-based central office. This will include four-wire AUTOVON telephone lines. Incoming and outgoing calls will be processed by the shore-based telephone system and the switching subsystem without the intervention of the SLCT attendant. In the event the terminal is busy, an alerting signal is provided as described in table X. If the busy terminal does not answer within 30 seconds, the calling party is disconnected and the trunk is made available for use.

3.6.7.2.4.1 Characteristics. Dedicated lines may be assigned to dial, automatic, and multiline terminals. Assigned dedicated lines and associated terminals shall be terminated in the same center.

3.6.7.2.4.2 Terminals to which dedicated lines are assigned will have all normal IVCS capabilities in addition to the dedicated line feature.

3.6.7.2.4.3 Incoming calls from the shore by dedicated lines shall be capable of being forwarded in the same manner as other calls if the terminal has the call forwarding permit.

3.6.7.3 Interface electronics module. The interface electronics module shall provide the interface circuitry necessary to allow IVCS dial terminals to place calls to, and receive calls from, shore telephone exchanges in the same manner in which internal IVCS calls are accomplished. There shall be one module for each shore line and each shall be capable of being configured to interface with both types of shore lines. Circuitry shall also be included to allow transmission of either dial pulse signals or DTMF when required by the shore exchange.

3.6.7.3.1 The interface electronics module shall perform the following functions as required by the type of shore equipment to be interfaced:

(a) A/D conversion.
(b) D/A conversion.
(c) Audio signal filtering.
(d) Transmit/receive signal amplification for supervisory signals and audio.
(e) Two-wire to four-wire conversion.
(f) Ring signal detection.
(g) Dial pulsing.
(h) Polarity detection.
(i) Shore line supervision.
(j) AUTOVON interface.

3.6.7.3.2 The signals received at the switching matrices from the interface electronics module shall be consistent with the signal characteristics received from other termination units. Signals from the interface electronics module to shore telephone exchanges shall meet the requirements of those systems.

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3.6.7.4 Shore line control terminal. The SLCT provides the attendant with the means of processing calls in the manual mode between IVCS subscribers and shore-based subscribers. One R/S from each switching center shall be dedicated by software to the SLCT while in port. The terminal shall perform the following functions:

(a) Originate and answer calls from IVCS subscribers and shore-based subscribers.
(b) Issue line HOLD commands.
(c) Issue release commands to release terminal switching center appearances for new calls.
(d) Provide an indication of calls in queue awaiting service.
(e) Provide visual and audible indications of incoming calls.
(f) Provide the capability for the attendant to talk to ship only, shore only, or in the three-way talk mode.

3.6.7.4.1 The SLCT shall utilize, to the maximum extent possible, the same electronics module and circuitry as the multiline terminal. Any differences between the two shall be accommodated by panel wiring and programming switches or jumpers.

3.6.8 Navy tactical data interphone system (NTDIS) interface. The NTDIS interface shall be in accordance with RCA Drawing 856-0003.

3.6.9 Ship's system interfaces.

3.6.9.1 Description. The IVCS shall interface with the man-on-the-move (MOM) system, general announcing system, and the sound-powered telephone system.

3.6.9.2 Performance. The electrical and physical interfaces between the IVCS and the ship's systems are identified in 3.6.9.3 through 3.6.9.5. These interfaces shall not degrade the IVCS transmission requirements specified in 3.5.2.4. The interfaces shall also accommodate the performance specifications of the ship's systems identified in 3.6.9.3 through 3.6.9.5.

3.6.9.3 Man-on-the-move (MOM).

3.6.9.3.1 Description. The MOM is a radio communication system that provides radio communication between personnel equipped with portable transmitter-receiver units by a multichannel base station. Each base station channel will be on a different frequency, permitting all MOM system nets to be in operation simultaneously.

3.6.9.3.2 Performance. The IVCS shall provide the means for designated dial and net terminals to enter MOM nets through the base station. Radio interface units (RIUs) in the switching center shall be utilized to interface with MOM base station channels. An RIU shall provide two- to four-wire conversion, transmitter PTT control, and impedance matching between the switching center and MOM base stations. There shall be one matrix appearance for each RIU and the RIU shall open to full duplex.

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3.6.9.3.2.1 The number of MOM nets interfaced shall be determined by the number of RIUs installed as specified (see 6.2.1). In configurations of dual switching centers, the MOM RIUs will be equally distributed between the centers. Access to an RIU from a remote switching center shall be by dedicated trunks to provide nonblocking service.

3.6.9.3.3 Interface requirements. Access to the RIUs shall be programmable to allow single or multiple access. When programmed for single access, authorized subscribers will get a busy tone if the RIU is already in use. If programmed for multiple access, its operation will be similar to that of a net, for example, any authorized subscriber calling the radio will be connected and disconnected without disturbing the remaining subscribers. In both cases, subscribers without authorization to the radio access will get an unavailable tone. Override of a radio call shall not be provided.

3.6.9.3.3.1 All RIUs shall have the following transmit or receive electrical characteristics:

(a)	Transmit	
	Output impedance: (floating/balanced)	600 ohms ± 10 percent
	Load impedance:	600 ohms ± 10 percent
	Audio output power:	1 milliwatt (0 dBm)
	Transmitter keying signal:	This shall consist of a relay contact closure on the RIU. Both relay contacts shall be "floating".
(b)	Receive	
	Input impedance: (floating/balanced)	600 ohms ± 10 percent
	Audio input power:	1 milliwatt (0 dBm)

3.6.9.4 General announcing system.

3.6.9.4.1 Description. The general announcing systems are used to pass orders, information, and alarms simultaneously to the entire ship or to limited areas.

3.6.9.4.2 Performance. Designated dial terminals shall have access to the general announcing system via the IVCS. An announcement may be initiated from a designated dial terminal by dialing the desired announcing system access code. The user will then be connected to the desired interface termination. Operation of the PTT switch shall cause a control signal to be sent to the announcing system amplifier changing it from standby to ready condition. The announcing system terminations will consist of amplifiers and associated loudspeaker groups, along with necessary control circuitry. The number of interface units will be specified (see 6.2.1). When dual centers are configured these interfaces shall be equally distributed between the switching centers.

3.6.9.4.3 Interface requirements. The announcing system interface lines shall be controlled.

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3.6.9.4.3.1 The interface lines shall appear as line appearances in the matrix of the IVCS switching center to which they are connected. Access to an interface from the remote center shall be via trunks.

3.6.9.4.3.2 Only one terminal shall be connected to a particular announcing system termination at any one time.

3.6.9.4.3.4 If microphone controls are provided, the use of the announcing system by a microphone control station shall cause a contact closure signal to be sent to the associated interface to indicate to the switching center that that portion of the announcing system is in use.

3.6.9.4.3.5 When a predesignated dial terminal accesses the general announcing system, the switching center shall send a contact closure to the announcing system to activate the local speaker muting feature of that system. The contact closure shall be capable of switching 28 V and 0.5 amps. The number terminals which will have the capability for local speaker muting to avoid feedback will be specified (see 6.2.1). The general announcing interface shall have the same send electrical characteristics as listed for RIUs in 3.6.9.3.3(a).

3.6.9.5 Sound-powered telephone system.

3.6.9.5.1 Description. The sound-powered telephone systems provide supplementary and auxiliary circuits for various subordinate control, operating, and service functions. They also provide an alternate means of communication for specific nets within the IVCS in the event of casualty to those circuits.

3.6.9.5.2 Performance. The number of sound power interface units for each center shall be specified (see 6.2.1).

3.6.9.5.3 Interface requirements. Each interface unit shall be connected to a net on the IVCS side of the interface. Each interface shall provide impedance matching and electrical isolation across the interface and shall be capable of operating with six standard Navy sound-powered phones in accordance with MIL-T-15514.

3.7 Major component list. The IVCS consists of the components listed in table XIII. The quantity of each component for a specific installation will be specified (see 6.2.1).

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TABLE XIII. Major components.

Component	Paragraph
Power supply and power distribution subsystem	3.6.1
D.c. power breaker panel	3.6.1.4
D.c. power supply	3.6.1.5
Batteries and rack	3.6.1.6
Control subsystem	3.6.2
Processors	3.6.2.3
Switchover control unit	3.6.2.4
Memories	3.6.2.5
Control programs	3.6.2.6
Switching subsystem	3.6.3
Switching matrices	3.6.3.3
Termination units	3.6.3.4
Shore line interface	3.6.7
Interface electronics module	3.6.7.3
Shore line control terminal	3.6.7.4
Distribution subsystem	3.6.4
Main distribution frame	3.6.4.3
Interrack cabling	3.6.4.4
Net switching matrix	3.6.4.5
Peripherals	3.6.5
Magnetic tape cartridge unit	3.6.5.3
Line printer	3.6.5.4
Terminals and accessories	3.6.6

3.8 Physical characteristics.

3.8.1 Weight.

3.8.1.1 Maximum weight. The IVCS shall be constructed to minimize the weight of the system.

3.8.1.2 Lateral stability. The center of gravity (CG) of the major deck- or desk-mounted component assemblages shall be such that the angle between a vertical line (90 degrees to the base line) through the CG and a line through the CG to the nearest outer base shall be greater than 25 degrees (see 4.5.3.1).

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3.8.1.3 Weight distribution. Weight distribution shall be such that an approximately even load is placed on the bottom corners of deck-mounted cabinets. Any lateral unbalance shall not exceed 10 percent of the gross weight of the major component assemblage. Port, starboard, forward, and after unbalance shall also not exceed 10 percent of the major component assemblage weight (see 4.5.3.1).

3.8.2 Dimensions. Major components shall be designed to allow installation in standard 19-inch equipment cabinets (19 inches wide, 26 inches deep, 72 inches high). The overall width dimension shall not exceed 24 inches. The dimensions include an additional constraint that the design for cable entry shall not require more than 1 foot of space above and below equipment cabinets (see 4.5.3.2).

3.8.3 Environmental control requirements. The IVCS shall not require the provision of any special environmental control mechanisms and will operate under the conditions specified in 3.8.9 (see 4.5.3.3).

3.8.4 External color and finish. The contractor shall provide painting in accordance with MIL-E-16400. Finish coats for switching center enclosures shall be as follows:

(a) The casting and nonfunctional areas shall be light gray, formula No. 11, type III, class 2, semigloss in accordance with MIL-E-15090.
(b) The color of the functional control area (panel) shall be in accordance with MIL-P-7788 and FED-STD-595, color No. 37038 (black).

3.8.5 Enclosures. The equipment enclosures shall be in accordance with requirement 55 of MIL-STD-454 and MIL-E-16400.

3.8.6 Availability. The IVCS shall meet an intrinsic availability (A_i) requirement of .999995 using the following formula:

$$A_i = \frac{MTBF}{MTBF + MTTR}$$

MTBF — Mean time between failures (MTBF)

MTTR — Mean time to repair (MTTR)

where a system failure is defined as the condition existing when 15 percent or more of the terminals served by a switching center do not have full service (for example, the ability to promptly and successfully originate or receive calls and use all permits associated with the particular terminals).

3.8.7 Reliability/maintainability (R/M) program. An effective R/M program shall be developed in accordance with the data ordering document specified in the contract or purchase order (see 6.2.2). In addition, NAVSHIPS 0967-LP-312-8010 and NAVSHIPS 0967-LP-316-8010 shall be used as guides.

3.8.7.1 Reliability program planning. A reliability program shall be developed in accordance with the data ordering document specified in the contract or purchase order (see 6.2.2).

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3.8.7.2 Reliability requirements. The MTBF requirements for each equipment, as defined herein, shall not be less than the hours indicated in table XIV at the normal operating environmental conditions.

TABLE XIV. Component reliability.

Component	Specified MTBF (hours)	
	25°C	55°C
D.c. power breaker panel	100,000	100,000
A.c. or d.c. power supplies	15,000	10,000
Processor	12,500	10,000
Switchover control unit	47,000	38,000
Memory - 32K	3,000	1,000
Switching matrix	10,500	10,000
Termination units (line, net, trunk, and interface)	50,000	30,000
Interface electronics modules	41,000	30,000
Magnetic tape cartridge unit with power supply	3,000	2,000
Line printer	5,000	3,000
Shore line control terminal	20,000	10,000

3.8.7.3 MTBF prediction. A reliability prediction shall be performed for each end item cited in the contract in accordance with the part stress analysis prediction procedure of MIL-HDBK-217 (see 4.5.5.3). The prediction shall assume a shipboard sheltered environment. The contractor shall evaluate the prediction with respect to the reliability required for a production unit. If there is a discrepancy between the predicted MTBF and required MTBF, the contractor shall identify the cause of the discrepancies and determine the actions required to assure that the specified values are achieved.

3.8.7.4 Parts reliability program. A parts selection and application program shall be conducted to assure the use of reliable parts and the development of reliable equipments. NAVSEA 0967-LP-597-1011 shall be used as a guide (see table XV).

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TABLE XV. Part derating requirements.

Part type	Mil-spec (style)	Electrical		Maximum ambient temperature (°C)
		Parameter	Maximum percent of rated electrical stress	
Resistors				
Fixed Carbon Composition	MIL-R-329008 (RCR)	Power/voltage	50/80	120 for RCR05, 07 - 100 for RCR20, 32, 42
Film	MIL-R-11804 (RD)	Power/voltage	50/80	200
	MIL-R-39017 (RLR)	Power/voltage	50/80	130
	MIL-R-55182 (RNR)	Power/voltage	50/80	155
	MIL-R-83401	Power/voltage	50/80	105
Wirewound	MIL-R-26 (RW)	Power/voltage	50/80	200
	MIL-R-18546 (RE)	Power/voltage	50/80	185
	MIL-R-39005 (RBR)	Power/voltage	50/80	125
	MIL-R-39007 (RWR)	Power/voltage	50/80	205
	MIL-R-39009 (RER)	Power/voltage	50/80	185
Variable				
Composition	MIL-R-94 (RV)	Power/voltage	50/80	100
Wirewound	MIL-R-19 (RA)	Power/voltage	50/80	85
	MIL-R-22 (RP)	Power/voltage	50/80	320 for RP05, 06 10, 15, 20 and 25 370 for RP30
	MIL-R-12934 (RR)	Power/voltage	50/80	130
	MIL-R-27208 (RT)	Power/voltage	50/80	130
	MIL-R-39002	Power/voltage	50/80	115
	MIL-R-39015 (RTR)	Power/voltage	50/80	130

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TABLE XV. Part derating requirements. - Continued

Part type	Mil-spec (style)	Electrical		Maximum ambient temperature (°C)
		Parameter	Maximum percent of rated electrical stress	
Nonwirewound	MIL-R-22097 (RJ)	Power/voltage	50/80	130
	MIL-R-23285 (RVC)	Power/voltage	50/80	155
	MIL-R-39023 (RQ)	Power/voltage	50/80	105
	MIL-R-39035 (RJR)	Power/voltage	50/80	120
Capacitors mica	MIL-C-5 (CM)	Voltage	60 percent dipped	135 for all except CM35
			40 percent molded	105 for CM35
	MIL-C-10950 (CB)		60 percent dipped	130 for all except CB50
			40 percent molded	65 for CB50
	MIL-C-39001 (CB)		50	105
Glass	MIL-C-14409 (PC) (Variable)	Voltage	50	105 for PC25, 26, 39, 43, 48 and 42, 130 for others
	MIL-C-23269 (CYR)		50	100
Electrolytic	MIL-C-39003 (CSR)	Voltage	50	65 (increased to 105 with further derating)
	MIL-C-39006 (CLR)	Voltage	60	65 (increased to 105 with further derating)
	MIL-C-39018 (CU)	Voltage	80	65 (increased to 105 with further derating)
	MIL-C-55365 (CWR)	Voltage	50	105

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TABLE XV. Part derating requirements. - Continued

Part type	Mil-spec (style)	Electrical		Maximum ambient temperature (°C)
		Parameter	Maximum percent of rated electrical stress	
Paper	MIL-C-12889 (CA)	Voltage	50	65
	MIL-C-39022 (CHR)	Voltage	50	65 for CHR09; 105 for CHR19
Paper plastic	MIL-C-19978 (CQR)	Voltage	50	105 for CQR07, 09, 12 and 13; 65 for CQ29, 32, 13
Ceramic	MIL-C-20 (CCR)	Voltage	50	105
	MIL-C-81 (CV) (Variable)	Voltage	50	65
	MIL-C-11015	Voltage	50	65, 105, or 130 depending on the capacitor being used (based on Tmax -200°C)
	MIL-C-39014 (CKR)	Voltage	50	65 for CKR64 and CKR72; 105 for others
	MIL-C-55681 (CDR)	Voltage	50	105
Gas or vacuum	MIL-C-23183 (CG) (Both fixed and variable)	Voltage	50	65
Semiconductors diodes	MIL-S-19500	Power	50	For additional information see section 300
		Forward continuous current	50	$\frac{T_s + T_{max}}{2}$
		Inverse voltage	65	where
		Transient voltage	80	T_s = max oper temp at full stress

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TABLE XV. Part derating requirements. - Continued

Part type	Mil-spec (style)	Electrical		Maximum ambient temperature (°C)
		Parameter	Maximum percent of rated electrical stress	
Semiconductors (continued) diodes		Surge current	70	T_{max} = Max_{junc} temp from spec
Transistors	MIL-S-19500	Power	50	
		Forward continuous	50	Same as for diodes
		Forward surge	75 (or 90 percent of the rated continuous current)	
		Voltage	75 across any junction	
		Reverse junction voltage (signal)	65 across any junction	
		Reverse junction voltage (surge)	75 across any junction	
Connectors	All those listed in MIL-STD-1353	Pin current voltage	50 25 percent of the dielectric withstanding voltage	20 degrees less than the specified maximum
Relays	All those listed in MIL-STD-1346	Contact current (continuous	75 - capacitive load 75 - resistive load 40 - inductive load 20 - motor 10 - filament	20 degrees less than the specified maximum
		Contact current (surge)	80	
		Contact power	25	
		Coil energize voltage	90 minimum	

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3.8.7.4.1 Derating. Electrical or electronic parts shall not be electrically stressed greater than the listed percentage of rated value (see table XV) at maximum environmental stress, unless approved by the Naval Sea Systems Command (NAVSEA).

3.8.7.4.2 Parts selection. Electrical or electronic parts shall be in accordance with the following minimum requirements:

(a) Microelectronic parts shall be in accordance with all the requirements of MIL-M-38510, class B.
(b) Discrete semiconductor parts shall be in accordance with all the requirements of MIL-S-19500, JANTX type. The maximum ratings as determined by the safe operating area (SOA) curves for power-switching transistors shall not be exceeded.
(c) Electrical or electronic parts shall meet the NAVSEA 0967-LP-597-1011 established requirements of reliability of 0.1 percent failure/1000 hours (P level quality).
(d) Microelectronic parts screened, burned-in, tested to the class B product assurance requirements of MIL-M-38510 may be used in the equipment only if justified and approved by NAVSEA. These parts are nonstandard parts and must be marked with the contractor's part number.
(e) Semiconductor parts screened, burned-in and tested to the TX quality assurance provisions of MIL-S-19500 may be used in the equipment only if justified and approved by NAVSEA.
(f) The use of other electrical or electronic military standard parts shall be allowed only if justified and approved by NAVSEA.

3.8.7.4.3 Parts control program. The parts control program shall be done in accordance with the data ordering document specified in the contract or purchase order (see 6.2.2).

3.8.7.4.4 Reliability test requirements.

3.8.7.4.4.1 Operational burn-in test. An operational burn-in test shall be conducted as specified in 4.5.5.

3.8.7.4.4.2 Reliability qualification test. The reliability qualification test shall be conducted as specified in 4.5.5.3.

3.8.7.4.5 Failure mode and effect analysis (FMEA). A failure mode and effect analysis shall be done in accordance with the data ordering document specified in the contract or ordering document (see 6.2.2).

3.8.7.4.6 Electrostatic discharge (ESD) protection requirements. When metal oxide semiconductor (MOS) parts and other similar parts sensitive to ESD are utilized in the equipments, it shall be ensured that ESD sensitive parts or subassemblies are protected in all phases of handling and testing, affix warning labels to protective packaging and to the equipment, provide warnings in all relevant areas of the equipment technical manual, and use identification marking on all ESD sensitive subassemblies visible to maintenance personnel (for example, on

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card edges) prior to maintenance handling of the equipment. Spare parts, modules, PWAs or subassemblies shall also be protected from ESD damage. DOD-STD-1686 and DOD-HDBK-263 shall be utilized as guidance for the identification and the protection of sensitive electronic components and assemblies.

3.8.7.5 Maintainability requirements.

3.8.7.5.1 Maintainability prediction. The maintainability prediction shall be prepared in accordance with the data ordering document specified in the contract or purchase order (see 6.2.2).

3.8.7.5.2 Corrective maintenance requirements. Corrective maintenance time does not include time to obtain replacement items, but shall include: preparation time (test analysis of fault information), fault localization time, fault isolation time, warm-up time, and repair validation time. Repair times shall be based on all maintenance tasks being performed by one individual with USN basic IC skills and system maintenance training. The organizational corrective maintenance MTTR for the equipment when configured as a system shall not exceed 1-1/2 hours. The organizational maximum repair time at the 95th percentile shall not exceed 3 hours as listed below:

Component	Apportioned MTTR
D.c. power breaker panel	0.5
A.c. or d.c. power supplies	0.5
Processor	1.0
SCU	0.25
Memory - 32K	0.5
Switching matrix	0.5
Termination units (line, net, trunk, and interface)	0.25
Interface electronics modules	0.5
Magnetic tape cartridge unit with power supply	1.0
Line printer	1.0
SLCT	1.0

3.8.7.5.3 Planned maintenance requirements. Periodic scheduled maintenance requirements shall be consistent with the MTBF figures and the requirements specified in 3.8.8.5. Scheduled maintenance shall include, but not be limited to, air filter cleaning and replacement, battery cleanliness and specific gravity checks, daily semiautomatic system dial-in tests from the system maintenance terminal, lamp and meter checks, and general cleanliness requirements.

3.8.7.5.4 Built-in test equipment (BITE). The equipment on-line performance monitoring and operational requirements, and the maintainability requirements shall be analyzed to develop the equipment BITE concepts as follows:

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(a) Automated fault detection and location shall be provided down to a single line replaceable unit (LRU), as a goal, but not more than three LRUs. The degree of automation through special hardware devoted exclusively to this function, test routines, and procedures described in the technical manuals shall be such that the system MTTR is achieved. Fully automated detection of faults will not be provided for terminals or SLCT although detailed fault isolating procedures will be provided in technical manuals. Memory errors will be isolated to the specific word or a small group of words. Error checking codes will be utilized to aid in this capability.

(b) Redundant equipments which are not utilized during the normal operating cycles of the system shall be periodically checked by semiautomatic maintenance routines.

(c) Maintenance testing or repairs shall normally be accomplished while the system is on line. The LRUs may be replaced, memory words revised, or others, without shutting the system down; however, the replaced equipment shall be off when repair is performed.

(d) Alarms and displays shall be provided for alerting maintenance personnel to the occurrence of a switching center fault. Switching center alarms shall be remote from the other switching center in a dual configuration. These alarms shall be provided, such that differentiation between major and minor failure is readily apparent. The displays shall indicate the nature of the fault. Manual means for resetting the alarms shall be provided.

(e) Automatic printout equipment shall be activated in response to requests for special routines from the maintenance terminal to verify requests and to provide results. It shall also be automatically activated in cases of stored program, fault detecting routines to provide an indication of detected faults.

(f) The maintenance circuitry displays shall be physically located entirely on the front panel of the equipment cabinets.

(g) Means for manually testing the maintenance circuitry shall be provided.

(h) Access to test points, replaceable modules, and limited-life components shall be simple and direct.

(i) Automatic routine maintenance operations will be performed on a not-to-interfere basis with system traffic.

(j) The tests to be performed automatically shall include checking of common subsystems, such as the processor, memory and matrix operation, and any other portions of the system deemed necessary to meet maintainability requirements.

(k) The pair of wires that supply power to individual terminal units shall be fused on one side at the switching center. An audible alarm shall be provided to indicate fuse failure. Visual indications will be provided on the fuse panel to locate the blown fuse.

(l) Manually initiated automatic tests of the circuitry shall include: continuity tests of each net circuit, tests for short circuits across the audio pair of each line with net access, continuity of transmission paths, line circuit operation, trunk circuit operation, and short circuits across the audio pair of each line terminated in the matrix.

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(m) A disconnect facility shall be provided to permit rapid isolation of any faulty or damaged wiring leading to a net terminal which, due to the nature of the fault (such as grounds or short circuits), causes degradation to the overall performance of a switching center.
(n) Provisions shall be made to permit maintenance personnel to make tests of individual telephone lines readily and easily at the switching center.

3.8.7.5.4.1 The technical manuals provided shall include any additional information required to isolate and repair a fault, including a dictionary which relates to the printout codes and displays provided in accordance with the data ordering document specified in the contract or purchase order (see 6.2.2).

3.8.7.5.5 Maintainability demonstration. A maintainability demonstration shall be conducted as specified in 4.5.5.2.

3.8.8 Electrical and mechanical design. The mechanical and electrical design shall be as specified herein.

3.8.8.1 Accessibility. Accessibility for repair and replacement of subassemblies and parts shall be in accordance with MIL-E-16400. The equipment cabinet doors shall be free to swing in an arc of 130 degrees with latching mechanisms placed at 90 and 130 degrees for maintaining the open position. Additionally, removable access panels shall be provided at the back of each equipment cabinet.

3.8.8.2 Test points. Test points for the connection of manual and automatic test equipment shall be provided within each switching center. As a minimum, these test points shall include:

(a) Test points at the switching matrix to allow verification of switching center terminations. This point shall be in addition to the points required to provide normal switching capabilities.
(b) The ability to connect either processor to determine its operational integrity independent of other switching center subsystems.
(c) Test points as required to verify the performance of other switched elements, switching center peripherals, and SLCT.
(d) Test points positioned on all equipment down to the lowest replacement level to permit fault isolation of individual piece parts.
(e) Printed wiring assembly configurations shall ensure compatability with software-programmed, depot-level, diagnostic test equipment. Emphasis is to be placed on the design of the PWAs for testability and repair. Test points located within the PWAs shall be in accordance with MIL-STD-415, class D test provisions. Set, reset, inhibit, and other control lines within the PWA that facilitate testing shall be conveniently accessible and permit the injection and readout of test signals. Feedback loops within the PWA shall be externally accessible and permit opening or closing of the loops at strategic points.

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3.8.8.3 Calibration, alinement and adjustment requirements. Equipment components shall not require calibration, alinement and adjustment at the organizational repair level.

3.8.8.4 Design for maintenance. The design for maintenance shall be as specified herein.

3.8.8.4.1 Organizational level. Maintenance of the system shall require component or subassembly removal and replacement to be performed at the organizational level by Navy enlisted personnel who have satisfactorily completed a Navy class "C" school training course in IVCS maintenance.

3.8.8.4.2 Intermediate and depot level. For the failures occurring at the organizational level that require intermediate or depot maintenance, the MTTR of the failed assembly or subassembly shall be no greater than 2 hours.

3.8.8.5 Maintenance frequency (scheduled and nonscheduled). The switching center shall operate for extended periods without the need for intervening corrective maintenance by organizational level personnel. The system shall be designed such that scheduled maintenance can be performed on line and organizational level maintenance personnel ordinarily shall be required to attend to the switch on a scheduled basis only. Scheduled maintenance shall occur once every 24 hours. The total 24-hour maintenance burden, scheduled and nonscheduled, for a switching center shall not average more than 2 man-hours a day. The probability of exceeding 24 man-hours a day shall be less than 0.01.

3.8.9 Environmental conditions. The equipment shall neither show degradation of performance specified in 3.2, 3.3, 3.4, and 3.5; nor malfunction of equipment; nor shall parts or electrical connections become loose, break, or go out of adjustment when subjected to the environment specified herein.

3.8.9.1 Dust tightness. The equipment shall be designed to be in accordance with the dust tightness requirement of MIL-E-16400.

3.8.9.2 Inclination. The inclination requirement shall be in accordance with MIL-E-16400.

3.8.9.3 Dripproof. The dripproof requirement shall be "dripproof (45 degrees)" in accordance with MIL-STD-108.

3.8.9.4 Salt spray. The salt spray requirement shall be in accordance with MIL-E-16400.

3.8.9.5 Accelerated life. The accelerated life requirement shall be in accordance with MIL-E-16400.

3.8.9.6 Vibration. The vibration requirement shall be in accordance with the type I requirement of MIL-STD-167-1. Calls shall neither be established nor lost while being subjected to this test.

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3.8.9.7 Shock. The H.I. shock requirement will be grade A, class I, type A in accordance with MIL-S-901.

3.8.9.8 Temperature extremes. IVCS equipment shall meet the temperature requirements in accordance with MIL-E-16400, except where indicated herein, for the specified degree of equipment exposure. Common control and switching equipment with associated control and I/O devices shall be defined as sheltered equipment. Terminal equipment, including telephone instruments, speakers and headsets installed within ships spaces, shall be defined as exposed equipment. Topside terminal equipment in their associated watertight enclosures shall be defined as exposed, topside equipment. Equipment temperature requirements shall be as specified in 3.8.9.8.1 and 3.8.9.8.2.

3.8.9.8.1 Operating. The minimum operating temperature shall be 0°C and the maximum 55°C for all IVCS sheltered equipment and -28°C to a maximum of 65°C for all IVCS exposed, unsheltered equipment.

3.8.9.8.2 Extended storage. The IVCS equipment, including storage batteries, shall meet the requirements of MIL-E-16400. Storage batteries shall meet this requirement only in the charged and dry state.

3.8.9.9 Humidity. The equipment shall maintain the specified performance when exposed to a relative humidity of 95 percent for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the equipment in the form of either water or frost.

3.8.9.10 Fungus. The IVCS equipment shall be resistant to fungi and shall not be adversely affected by fungi under conditions of high humidity, temperature, and presence of inorganic salt.

3.9 Design and construction. The IVCS shall meet the requirements of MIL-E-16400 and to the extent specified herein.

3.9.1 Materials and parts.

3.9.1.1 Materials. The selection and use of materials shall be in accordance with MIL-E-16400.

3.9.1.2 Parts. Parts requirements shall be in accordance with MIL-E-16400. The following requirements shall apply to shielded wires:

(a) Single shielded wire. When only one shielding wrap is used for single or multiple conductor wire or coaxial cable, the shield shall be electrically insulated from all signal conductors; and the shield shall also be electrically insulated from ground by means of an integrally molded fire retardant insulating wrap.
(b) Multiple shields. Multiple shields are described as the use of two or more concentrically wrapped shields. Shields shall be electrically insulated from each other.
(c) Armor. When a conducting armor covering is used to mechanically protect the shielded wires, the armor shall be electrically isolated from all conductors, shields, and ground by means of an integrally molded and approved fire retardant insulating wrap.

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(d) Power supply returns. No shield or armor covering shall be used as a conductor to carry any power supply current (ground return or "live" side of circuit), whether the power supply is a local module, battery, or the ship's a.c. or d.c. supply line. The armor covering shall only be used for mechanical protection of the interior shielded wires. The shields shall only be used for electrical isolation of the signals in the inner conductors or for carrying signal return currents.

(e) Grounding. Circuit shielding shall be positively connected (bonded) to the unit enclosure or the chassis. Power supply returns shall be positively connected (bonded) to ship's hull at one (physical) point only. The ship's hull, internal chassis, or unit enclosure shall not be used in lieu of appropriate electric conductors in equipment circuitry unless specifically approved by NAVSEA for each application. No electric circuit which connects directly to ship's wiring external to the unit enclosure shall be electrically connected to the ship's hull or placed at ship's hull potential. Each switching center shall have a common tie point for signal ground. The common points of each center, in a dual center configuration, shall be tied together by the d.c. power supply common return lines. Provisions shall be available in the switching center to ground the common signal ground to the ship's hull at a single point.

3.9.1.2.1 Conformal coating. The PWAs will be conformal coated with self-healing silicone and shall be in accordance with MIL-STD-454 and MIL-I-46058.

3.9.2 Electromagnetic environment.

3.9.2.1 Electromagnetic pulse (EMP). The IVCS as installed aboard ship shall be able to operate in an environment which includes periodic broadband EMP of 50,000 volts per meter at exposed terminals with 50 volts per meter present inside the terminals and switching equipment. Such an EMP will not cause loss of service except to exposed terminals.

3.9.2.2 Electromagnetic radiation. The IVCS equipment and component subsystems shall be in accordance with the emission requirements of MIL-STD-461. The IVCS shall neither cause nor exhibit undesirable responses, malfunctions, or degraded performance when tested in accordance with MIL-STD-462. System compatibility shall be in accordance with MIL-E-6051 (see 4.5.7.3).

3.9.2.3 Bonding and grounding. Unless otherwise specified herein, all bonding and grounding shall be in accordance with MIL-STD-1310 (see 4.5.7.3.1).

3.9.2.4 Cables and connectors. Coaxial cables, if used, shall be of the double shielded type. Other cables requiring shields shall have a minimum of 90 percent coverage. Cable connectors shall be an approved type and provided with back shells for fastening the cable shield. The shield shall be constructed of braided, tinned copper wire. Shielded cables shall have the shield terminated in an approved military connector, having a radio frequency conductive finish and employing an electromagnetic interference back shell which provides

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for peripheral bonding of the shield. The use of shielded cable shall be kept to a minimum by the use of alternative suppression techniques. Panel mounted connectors and component surfaces shall be free of nonconductive finishes, to provide positive bonding and grounding with mating connectors and equipment ground plane. Cables and connectors shall be in accordance with MIL-STD-242 (see 4.5.7.3.2).

3.9.3 Nameplates and product marking. The nomenclature, identification markings, and labeling of major components and parts shall be accomplished in accordance with MIL-E-16400. Clean, concise, and durable markings shall be provided for all controls, lamps, switches, fuses, jacks, test points, and other components as necessary (see 4.5.7.4).

3.9.4 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9. Any special requirements or techniques shall be as specified by the Government (see 4.5.7.4).

3.9.5 Interchangeability.

3.9.5.1 Control programs. The software control programs shall be identical for all systems of the same capacity and completely interchangeable except for terminal access permits. The software shall be designed to accommodate the modular change in system capacity with minimum differences in the control programs (see 4.5.7.5).

3.9.5.2 Hardware. Hardware shall be in accordance with requirement 7 of MIL-STD-454.

3.9.6 Safety. Safety shall meet the requirements of MIL-E-16400 (see 4.5.7.6.1).

3.9.7 Human engineering. Human engineering design criteria and principles shall be applied in the design of the IVCS, so as to achieve safe, reliable, and effective performance by system operator or maintenance personnel and to minimize personnel skill requirements and training time. Unless otherwise specified herein (see 4.5.7.6), human engineering design criteria shall be in accordance with MIL-STD-1472.

3.9.7.1 Detail design. Detail design of the IVCS component equipments (including contractor acquired components) shall be in accordance with MIL-STD-1472.

3.9.8 TEMPEST. The design goal of all equipment shall be to produce a TEMPEST-secure system in accordance with the requirements of NACSEM 5100. Adequate separation and protection of circuits and lines shall be provided, including any necessary shielding and filtering conforming to the requirements of MIL-STD-1680. Crosstalk isolation between one audio pair and any other audio pair in the switch shall be greater than 83 dB with a design goal of 100 dB. Whenever possible, TEMPEST, EMI, and electromagnetic compatibility (EMC) requirements shall be considered simultaneously to minimize design cost. Signals at the IVCS internal and external interfaces shall be classified as black or red.

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3.10 Technical data.

3.10.1 Drawings. The contractor shall provide engineering drawings and associated lists in accordance with the data ordering document specified in the contract or order (see 6.2.2).

3.10.2 Technical manuals. The contractor shall provide technical manuals in accordance with the data ordering document specified in the contract or order (see 6.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract or purchase order (see 6.2.2).

4.2 Classification of inspections. The methods of inspection and test shall fall within the following classifications in accordance with MIL-E-16400 and listed in table XVI:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

(1) Production inspection (group A).
(2) Production control inspection (group B).
(3) Environmental inspection (group C).

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TABLE XVI. IVCS inspections and tests.

Subject	Requirement	Method	Test group A	Test group B	Test group C
Design and construction	3.9	4.5.7			X
Materials and parts	3.9.1	4.5.7.1			X
Electromagnetic pulse	3.9.2.1	4.5.7.2			X
Electromagnetic radiation	3.9.2.2	4.5.7.3			X
Bonding and grounding	3.9.2.3	4.5.7.3.1			X
Cables and connectors	3.9.2.4	4.5.7.3.2			X
Mechanical inspection	3.9.3	4.5.7.4			X
Workmanship	3.9.4				
Interchangeability	3.9.5	4.5.7.5			X
Safety	3.9.6	4.5.7.6.1			X
Physical characteristics	3.8	4.5.3	X		
Weight	3.8.1	4.5.3.1	X		
Dimensions/cube	3.8.2	4.5.3.2	X		
Environmental control	3.8.3	4.5.3.3			X
External finish	3.8.4	4.5.3.4	X		
Human engineering	3.9.7	4.5.7.6			X
Performance tests	3.5.2, 3.6	4.5.2	X		
Major components/subsystem performance	3.6	4.5.2.2	X		
Power supply and power distribution subsystem	3.6.1	4.5.2.1, 4.5.2.2	X		X
Control subsystem	3.6.2	4.5.2.1, 4.5.2.2	X		X
Switching subsystem	3.6.3	4.5.2.1, 4.5.2.2	X		X
Distribution subsystem	3.6.4	4.5.2.1, 4.5.2.2	X		X
Peripherals	3.6.5	4.5.2.1, 4.5.2.2	X		X
Shore line interface	3.6.7	4.5.2.1, 4.5.2.2	X		X
Navy tactical data interphone system (NTDIS) interface	3.6.8	4.5.2.1, 4.5.2.2	X		X
Ship's system interface	3.6.9	4.5.2.1, 4.5.2.2	X		X
System performance	3.5.2	4.5.2		X	X
Capacity	3.5.2.1	4.5.2.1, 4.5.2.3		X	X
Connectivity	3.5.2.2	4.5.2.1, 4.5.2.3			
Traffic handling	3.5.2.3	4.5.2.1, 4.5.2.3		X	X
Blocking	3.5.2.3.3	4.5.2.1, 4.5.2.3		X	X
Call initiation delay	3.5.2.3.4	4.5.2.1, 4.5.2.3		X	X
Call completion delay	3.5.2.3.5	4.5.2.1, 4.5.2.3			
Misrouting	3.5.2.3.6	4.5.2.1, 4.5.2.3		X	X
Transmission requirements	3.5.2.4	4.5.2.1, 4.5.2.3		X	X

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TABLE XVI. IVCS inspections and tests. - Continued

Subject	Requirement	Method	Test group A	Test group B	Test group C
TEMPEST tests	3.9.8				
Environmental	3.8.9	4.5.6			X
Inclination	3.8.9.2	4.5.6.1			X
Drip test	3.8.9.3	4.5.6.2			X
Salt spray	3.8.9.4	4.5.6.3			X
Accelerated life	3.8.9.5	4.5.6.4			X
Vibration	3.8.9.6	4.5.6.5		X	X
Shock	3.8.9.7	4.5.6.6			X
Temperature	3.8.9.8	4.5.6.7		X	X
Humidity	3.8.9.9	4.5.6.8			X
Fungus	3.8.9.10	4.5.6.9			X
Enclosures	3.8.5	4.5.6.10			X
RMA tests					
Availability	3.8.6	4.5.3			X
Reliability	3.8.7	4.5.4			X
Maintainability	3.8.7	4.5.4			X
Maintenance levels	3.8.8.4.1, 3.8.8.4.2				X
Maintenance frequency	3.8.8.5				X

4.3 Examination and test plan. A test plan shall be prepared in accordance with the data ordering document specified in the contract or purchase order (see 6.2.2).

4.3.1 Test reports. The contractor shall provide test reports in accordance with the data ordering document specified in the contract or purchase order (see 6.2.2).

4.3.2 Test facilities. The requirements for the test facilities utilized in the tests specified herein shall be in accordance with MIL-E-16400.

4.3.3 Test conditions. Unless otherwise specified herein, test conditions shall be in accordance with MIL-E-16400.

4.3.4 Performance of tests. Unless otherwise specified, the inspection and performance of tests shall be in accordance with MIL-E-16400. In the event of a failure of any tests specified herein or in MIL-E-16400, the entire test series shall be repeated after repair or replacement of the test item. Items previously qualified to the performance and environmental requirements of section 3 need not be requalified to be offered for delivery, subject to approval by NAVSEA.

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4.3.5 Order of tests. The tests shall be conducted in the order of their appearance in table XVI except that the order of the individual tests within each group may be varied subject to approval by NAVSEA.

4.4 First article.

4.4.1 First article inspection. The items designated for first article inspection shall be subjected to all of the examinations and tests listed in 4.6 and table XVII.

TABLE XVII. IVCS performance verification methods for first article.

Subject	Requirement	Test	Method	
			Analysis	Inspection
Major components - subsystem performance	3.6	X	X	X
Power supply and power distribution subsystem	3.6.1.2	X	X	
D.c. power breaker panel	3.6.1.4	X		X
D.c power supply	3.6.1.5		X	X
Batteries and racks	3.6.1.6	X	X	
Control subsystem	3.6.2.2	X		
Processor	3.6.2.3	X	X	
SCU	3.6.2.4	X		
Memory	3.6.2.5	X		
Control programs	3.6.2.6	X	X	
Switching subsystem	3.6.3	X	X	
Switching matrices	3.6.3.3	X		
Termination units	3.6.3.4	X		
Distribution subsystem	3.6.4.2	X		X
MDF	3.6.4.3	X		X
Interrack cabling	3.6.4.4	X		X
Net switching matrix	3.6.4.5	X		X
Peripherals	3.6.5	X		
Magnetic tape cartridge unit	3.6.5.3	X		
Line printer	3.6.5.4	X		
Maintenance terminal	3.6.5.5	X		
Shore line interface	3.6.7	X		X
Interface electronics module	3.6.7.3	X		
SLCT	3.6.7.4	X		

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TABLE XVII. IVCS performance verification methods for first article. - Continued

Subject	Requirement	Test	Method	
			Analysis	Inspection
Navy tactical data interphone system (NTDIS) interface	3.6.8	X		X
Ship's system interface	3.6.9	X		
MOM	3.6.9.3	X		X
General announcing	3.6.9.4, 3.6.9.4.3	X		X
Sound powered telephone system	3.6.9.5	X		X
System performance	3.5.2	X	X	X
Capacity	3.5.2.1			X
Connectivity	3.5.2.2		X	
Traffic handling	3.5.2.3		X	
Blocking	3.5.2.3.3		X	
Call initiation delay	3.5.2.3.4	X	X	
Call completion delay	3.5.2.3.5	X	X	
Misrouting	3.5.2.3.6		X	
Transmission requirements	3.5.2.4	X		

4.5 Quality conformance inspection.

4.5.1 Quality conformance inspections. Configuration and performance requirements for the IVCS shall be verified by analysis, inspection, or test. Verification of all requirements as specified in sections 3 and 5 shall be accomplished. The contractor shall perform the inspections and tests specified in this section and listed in table XVII. This does not relieve the contractor of his responsibility for performing any additional inspections that are necessary to control the quality of the items offered for delivery and to assure compliance with all requirements specified herein.

4.5.2 Performance tests. Performance tests shall consist of major component subsystem tests and system tests which are detailed herein.

4.5.2.1 Performance test configuration. The test configuration for performance tests shall consist of a complete system. It shall include interconnecting trunks, net interties, power and control cables, for dual center configurations. Test configurations shall include battery power supplies and a minimum of six of each type of terminal used with the ordered system. The test configuration shall be installed in a test facility in a manner that will simulate shipboard usage. Test instrumentation and connection to the test configuration shall be provided as necessary.

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4.5.2.2 Major component or subsystem performance verification. The performance of major components and subsystems shall be verified in accordance with MIL-E-16400 and the methods identified in table XVII.

4.5.2.3 System performance verification. The system performance specified in 3.5.2 shall be verified in accordance with MIL-E-16400 and the methods identified in table XVII.

4.5.3 Physical characteristics tests. The physical characteristics of the IVCS components as set forth in table XIV, including the enclosures, shall be verified by inspection and test in accordance with MIL-E-16400 and as follows.

4.5.3.1 Weight and dimensions. Maximum allowable weights and dimensions shall be verified by measurement. Compliance with the lateral stability requirement of 3.8.1.2 shall be verified by measurement of the CG location. Conformity to the weight distribution requirement of 3.8.1.3 shall be verified by analysis.

4.5.3.2 Dimension cube. The dimensions of major components shall be measured to verify compliance with 3.8.2.

4.5.3.3 Environmental control. Environmental control inspection shall be done to determine compliance with 3.8.3.

4.5.3.4 External color and finish. External color and finish shall be inspected to determine compliance with 3.8.4.

4.5.4 Availability test. The availability requirement specified in 3.8.6 shall be verified by an availability test prepared in accordance with the data ordering document specified in the contract or purchase order (see 6.2.2).

4.5.5 Reliability and maintainability tests.

4.5.5.1 Operational burn-in-test. An 100-hour operational burn-in test shall be performed on each equipment, PWA, and subassembly in accordance with MIL-STD-781 and the following environmental stresses:

(a) ON-OFF cycle: OFF 30 minutes each 24-hour period (cooling cycle).
(b) Electrical stress: Operational inputs shall be actual to maximum extent practicable. Input voltage cycling shall be 1/3 at nominal - 7 percent, 1/3 at nominal, 1/3 at nominal + 7 percent.
(c) Vibration stress: Random vibration shall be in accordance with the requirements of NAVMAT P-9492.
(d) Thermal stress: Temperature range, 0-50°C cycling.
(e) Rate of temperature change: Not less than 5°C per minute and not greater than 10°C per minute.
(f) Frequency of temperature change: Stabilization plus 2 hours minimum at high and low temperature.
(g) Moisture (humidity): Ambient.
(h) Duty cycle (modes of operation): Continuous.

The last 50 hours of the burn-in test for each equipment shall be failure free.

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4.5.5.2 Maintainability demonstration. The equipment shall undergo a maintainability demonstration (phase II) in accordance with MIL-STD-471, test method 9. The demonstration shall test organizational level corrective maintenance, preventive maintenance, and fault isolation requirements. The following conditions shall also govern the demonstration:

(a) Environmental constraints shall be simulated for actual shipboard installation.
(b) The tasks shall be selected in accordance with MIL-STD-471.
(c) The tasks shall require use of the equipment technical manual.
(d) Each preventive maintenance task shall be demonstrated three times.
(e) At least 60 corrective maintenance tasks shall be selected and demonstrated.

Should the demonstration detect equipment, software, or support item(s) deficiencies or result in a reject decision, retest is required. Total or partial retest shall be as determined by NAVSEA.

4.5.5.3 Reliability qualification test. A 2500-hour ON time reliability qualification test shall be performed in accordance with MIL-STD-781.

4.5.5.3.1 Test conditions and stresses. The test shall be performed in accordance with the following conditions and stresses:

(a) ON-OFF cycle: ON continuously.
(b) Electrical stress: Operational inputs shall be actual to the maximum extent practicable. Input voltage cycling shall be 1/3 at nominal, 1/3 at nominal $\pm$ 7 percent.
(c) Vibration stress: Swept-sine, continuous, sweep time 10 $\pm$ 2 minutes; amplitude, 0.020 $\pm$ 0.004 inches (DA); frequency range, 4 Hz - 33 Hz - 4 Hz.
(d) Thermal stress: Temperature range, 0-50°C cycling.
(e) Rate of temperature change: Not less than 5°C per minute and not greater than 10°C per minute.
(f) Frequency of temperature change: Stabilization plus 2 hours minimum at the high and low temperature.
(g) Moisture (humidity): Ambient.
(h) Operational mode is continuous.

Because equipment MTBFs are high, every effort shall be made to place the maximum quantities of equipment on test, for example, the higher the MTBF the greater the amount of those equipments on test.

4.5.5.3.2 Test accept or reject decision. Test accept or reject decision shall be as specified herein.

4.5.5.3.2.1 Acceptance shall be achieved if each equipment placed on test is operated for the required ON time test period, without system failure (see 3.8.6), and the recorded data verifies conformance with the requirements specified herein and the approved test plan or test procedures. Should part failures (not system failures) occur during testing, repair shall be performed and verified (by separate testing up to 8 hours), and testing shall continue.

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4.5.5.3.2.2 Should a part failure trend(s) develop within the first 100 ON time hours of test, the test shall be stopped, a failure analysis shall be performed, corrective action shall be taken, and the test shall be continued.

4.5.5.3.2.3 Should a part failure trend(s) develop after 100 ON time hours of test, repair shall be performed and verified (by testing up to 8 hours), and the test shall be continued. A failure analysis shall be performed and corrective action shall be implemented and verified. Corrective action verification requires a minimum of 50 ON time hours of testing. Failure actions shall be in accordance with MIL-STD-781. Should the test result in a reject decision (system failure), retest is required. Retest may be structured such that after corrective action has been implemented, the total system or one or more equipment types only may be retested as approved by NAVSEA.

4.5.6 Environmental tests. Environmental requirements specified in 3.8.9 shall be verified by the following tests.

4.5.6.1 Inclination test. Inclination requirements specified in 3.8.9.2 shall be verified by testing in accordance with MIL-E-16400.

4.5.6.2 Drip test. Dripproof requirements specified in 3.8.9.3 shall be verified by testing in accordance with MIL-STD-108.

4.5.6.3 Salt spray test. Salt spray requirements specified in 3.8.9.4 shall be verified by testing in accordance with MIL-E-16400 for exposed and sheltered equipment.

4.5.6.4 Accelerated life test. Accelerated life requirements specified in 3.8.9.5 shall be verified by testing in accordance with MIL-E-16400.

4.5.6.5 Vibration test. Vibration requirements specified in 3.8.9.6 shall be verified by testing in accordance with MIL-STD-167-1.

4.5.6.6 Shock test. Shock requirements specified in 3.8.9.7 shall be verified by testing in accordance with MIL-S-901 test procedures for type A tests on grade A, class I equipment.

4.5.6.7 Temperature tests. Temperature requirements specified in 3.8.9.8 shall be verified by testing in accordance with MIL-E-16400.

4.5.6.8 Humidity test. Humidity requirements specified in 3.8.9.9 shall be verified by testing in accordance with MIL-E-16400.

4.5.6.9 Fungus test. Fungus requirements specified in 3.8.9.10 shall be verified by testing critical samples in accordance with method 508 of MIL-STD-810.

4.5.6.10 Enclosures. Enclosure requirements specified in 3.8.5 shall be verified by testing in accordance with MIL-E-16400 and MIL-STD-108.

4.5.7 Design requirement verification. The configuration and construction requirements specified in 3.9 shall be verified as follows.

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4.5.7.1 Material, processes, and parts. Material, processes and parts requirements specified in 3.9.1 shall be verified by documentation in accordance with the data ordering document specified in the contract or purchase order (see 6.2.2).

4.5.7.2 Electromagnetic pulse (EMP).

4.5.7.3 Electromagnetic radiation (EMR) tests. Electromagnetic radiation requirements specified in 3.9.2.2 shall be verified by using the techniques specified in MIL-STD-462. Ambient electromagnetic levels of the test site shall be in accordance with MIL-STD-462 during testing, measured with the test sample deenergized, and shall be at least 6 dB below the allowable specified limit.

4.5.7.3.1 Bonding and grounding. Prior to performance of EMI/EMC testing, the bonding and grounding requirements specified in 3.9.2.3 shall be ascertained by visual or mechanical inspection.

4.5.7.3.2 Cables and connectors. Cables and connectors requirements specified in 3.9.2.4 shall be verified by inspection.

4.5.7.4 Mechanical inspection tests. The equipment shall be given a complete mechanical and visual inspection test to determine that the quality of all materials and workmanship is in compliance with the requirements of 3.9.3 and 3.9.4. Particular attention shall be given to the following:

(a) Completeness.
(b) Nameplates, identification markings, and labels.
(c) Operation of adjustable and sliding parts, thumb screws, controls, and switches.
(d) Finishes.
(e) Welded and soldered joints.
(f) Fit of components in their respective positions.
(g) Check of mounting devices.
(h) Lubrication and corrosion prevention.
(i) Safety features and interlocks.
(j) Fastening and securing devices or parts.
(k) Accessibility of components and parts for servicing.
(l) Cable between components, including plugs and receptacles.
(m) Grounding connections.
(n) Overall dimensions and weight check.
(o) Foreign objects, lint, and dust.
(p) Potential shorts and defective insulation.
(q) Other visual defects.

4.5.7.5 Interchangeability. The equipment shall be subjected to interchangeability tests in accordance with the provisions of a Government-approved test plan, designed and approved by NAVSEA, to verify compliance with the requirements specified in 3.9.5 and in accordance with MIL-STD-454.

4.5.7.6 Human engineering test. Human engineering testing shall be conducted in accordance with the provisions contained in a Government-approved test plan to ensure compliance with the requirements of MIL-STD-1472 and as specified herein (see 3.9.7). The tests shall demonstrate that the man-machine

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combination can accomplish the intended operation, control, and maintenance functions in accordance with system objectives and mission requirements tests shall be performed using personnel from the user population or one with equivalent technical skills. The tests shall be witnessed by cognizant Government personnel.

4.5.7.6.1 Safety. Inspection and demonstration tests of equipment assemblies shall be conducted to ensure compliance with the requirements of 3.9.6. Tests shall verify, as appropriate, the requirements of air transportation safety while in a transport mode, ground safety, system safety, personnel safety, radiation safety, and protection from dangerous materials and components.

4.5.7.6.2 Accessibility and operability of operating controls. The operating controls shall be tested under conditions similar to those of normal use, to include extreme low temperatures, to assure ready accessibility and operability of all knobs, dials, pushbuttons, switches, adjustable and sliding parts, thumb screws, connectors, controls and all other components used in the operation and maintenance of the IVCS.

4.6 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging and packing. Unless otherwise specified, preservation-packaging and packing shall be in accordance with MIL-E-17555 and the levels specified (see 6.2.1).

5.2 Marking. Packages and containers shall be marked in accordance with the requirements of MIL-STD-129. For level C preservation-packaging and packing unit, intermediate (when provided) and exterior packs shall, as a minimum, be marked as follows by any means which provides legibility and durability.

(a) National stock number (NSN) or part number when NSN is not available.
(b) Noun nomenclature.
(c) Quantity and unit of issue.
(d) Contract, purchase order, or delivery order number.
(e) Additional markings as may be required by the contract.
(f) Exterior shipping containers shall be marked with the appropriate address and caution marking except for full carloads or full truckloads to a single Continental United States (CONUS) consignee.
(g) One copy of the shipping documents and contract or order documentation shall be attached to the outside in an envelope or placed inside the container.

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(h) Department of Transportation (DOT) markings as applicable.
(i) Date of manufacturer, month, and year.
(j) Shelf life when required, shall be as specified in MIL-STD-129.

5.3 Cushioning, filler, dunnage, and wrapping materials.

5.3.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation/stowage.

5.3.2 Level C preservation-packaging and packing. When loose-fill materials are used for packaging and packing applications such as cushioning, filler, and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labeled with the following information:

"CAUTION

Contents cushioned with loose-fill material shall not be taken on board ship. Remove and discard loose-fill material. If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material."

5.3.3 Resistance to fire. Cushioning, filler, dunnage, and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire.

6. NOTES

6.1 Intended use.

6.1.1 General. The IVCS will be used in shipboard configurations to provide a voice distribution system capable of providing both administrative and tactical communications on a 24-hour basis. The IVCS shall accommodate the voice switching requirements of various ship classes providing switching services for 200 to 3200 switchable terminations. Particular configurations of switching centers will be used in the following general applications:

(a) Systems requiring 200 to 400 switchable terminations shall be configured as a single switching center.
(b) Systems requiring 400 to 1600 switchable terminations shall be configurable as either a single switching center or dual switching centers.
(c) Systems requiring 1600 to 3200 switchable terminations shall be configured as dual switching centers.

6.1.2 Operation. In dual center configurations, each switching center shall be functionally identical and capable of independent operation.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article sample, if required (see 3.1).
(c) Type and number of interconnections (see 3.5.2.2.1).
(d) Number of major components and peripherals (see 3.6.2, 3.6.5, 3.7).
(e) Number of magnetic tape cartridges containing processor control program (see 3.6.2.6 and 3.6.5.3).
(f) Type and number of permits, accesses, special functional features, and restrictions (see 3.6.2.6.3.4 and 3.6.2.6.3.8).
(g) Number of terminals to be restricted during line load control (see 3.6.2.6.3.5).
(h) Type and numbering of nets (including number of subscribers per net) (see 3.6.2.6.4) and accessories available (see 3.6.6.5) for use on an individual terminal (see table XII).
(i) Number of terminals with "busy party alternate" designation (see 3.6.2.6.3.9).
(j) Type of maintenance terminal (see 3.6.5.5).
(k) Type and number of dial terminations (see table XI) and performance and interface characteristics for terminals and accessories (see 3.6.6.2 and 3.6.6.5).
(l) Type and number of shore line interfaces (see 3.6.7).
(m) Type and number of interfaces with other systems (see 3.6.9).
(n) Level of preservation-packaging and level of packing required for shipment (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.6.2.6	Program package documentation	DI-S-2141	----
3.8.7	Reliability/maintainability program plan	DI-R-3533	----
3.8.7.1	Reliability program plan	DI-R-7079	----
3.8.7.4.3	Part control program plan	DI-E-7026	----
3.8.7.4.5	Failure mode, effects and criticality analysis plan	DI-R-7086	----
3.8.7.5.1	Maintainability prediction data	DI-R-5189	----

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.8.7.5.4.1 and 3.10.2	Technical manuals	DI-M-5200	----
3.10.1	Drawings, engineering and associated lists	DI-E-7031	Level 3 Design activity designation - Contractor Design activity drawing no. - Contractor
4.1.1	Inspection system program plan	DI-R-4803	----
4.3	Plan, master test and evaluation program	UDI-T-23943	----
4.3.1	Test and evaluation report	DI-T-5143	----
4.5.4	Systems/equipments availability analysis data	DI-R-5481	----
4.5.7.1	Report, materials and processes development and evaluation	UDI-S-21270	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 5830-N063)

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